

PRIMARY 4 MATHEMATICS SCHEME OF WORK

FIRST TERM

Week	Topic	Breakdown (Subtopics)
1	Whole Numbers (Part 1)	- Counting and writing numbers 1-1,000 - Place value: ones, tens, hundreds, thousands - Reading and writing numbers in words and figures - Expanded notation: $345 = 300 + 40 + 5$ - Comparing numbers using $>$, $<$, $=$ symbols - Ordering numbers in ascending and descending order
2	Whole Numbers (Part 2)	- Counting in 2s, 5s, 10s, 25s, 50s, 100s - Skip counting patterns and number sequences - Number patterns: identifying and continuing patterns - Odd and even numbers up to 1,000 - Rounding numbers to nearest 10 and 100 - Number line activities (0-1,000)
3	Whole Numbers (Part 3)	- Roman numerals: I, V, X, L, C, D, M - Reading and writing Roman numerals up to 100 - Converting between Roman numerals and Hindu-Arabic numerals - Factors and multiples (introduction) - Prime numbers (basic introduction: 2, 3, 5, 7, 11, 13) - Square numbers: $1^2, 2^2, 3^2, 4^2, 5^2$ (1, 4, 9, 16, 25)
4	Fractions (Part 1)	- Understanding fractions: parts of a whole - Identifying and naming fractions: $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$,

		$\frac{1}{5}, \frac{1}{8}, \frac{1}{10}$ • Numerator and denominator • Representing fractions using shapes and diagrams • Equivalent fractions: $\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$ • Comparing fractions with same denominator
5	MID-TERM EXAMINATION	• Assessment of Weeks 1-4
6	MID-TERM BREAK	• Holiday/Rest
7	Addition and Subtraction (Part 1)	• Addition of 2-digit and 3-digit numbers without regrouping • Addition with regrouping (carrying) • Addition of three or more numbers • Vertical and horizontal addition • Word problems on addition • Mental addition strategies
8	Addition and Subtraction (Part 2)	• Subtraction of 2-digit and 3-digit numbers without regrouping • Subtraction with regrouping (borrowing) • Subtraction across zeros: $500 - 237$ • Vertical and horizontal subtraction • Word problems on subtraction • Mental subtraction strategies
9	Addition and Subtraction (Part 3)	• Mixed addition and subtraction problems • Addition and subtraction of money (Naira and Kobo) • Checking subtraction using addition • Multi-step word problems • Real-life applications: shopping, saving, spending • Problem-solving strategies

10	Multiplication (Part 1)	• Understanding multiplication as repeated addition • Multiplication tables: 2, 3, 4, 5, 10 times tables • Commutative property: $3 \times 4 = 4 \times 3$ • Multiplication by 10, 100 (pattern recognition) • Multiplication of 2-digit by 1-digit numbers (without regrouping) • Word problems on multiplication
11	REVISION	• Review of all topics: Whole numbers, fractions, addition, subtraction, multiplication
12	EXAMINATION	• End of Term Assessment
13	CLOSING/VACATION	• Term ends



PRIMARY 4 MATHEMATICS LESSON NOTES

FIRST TERM

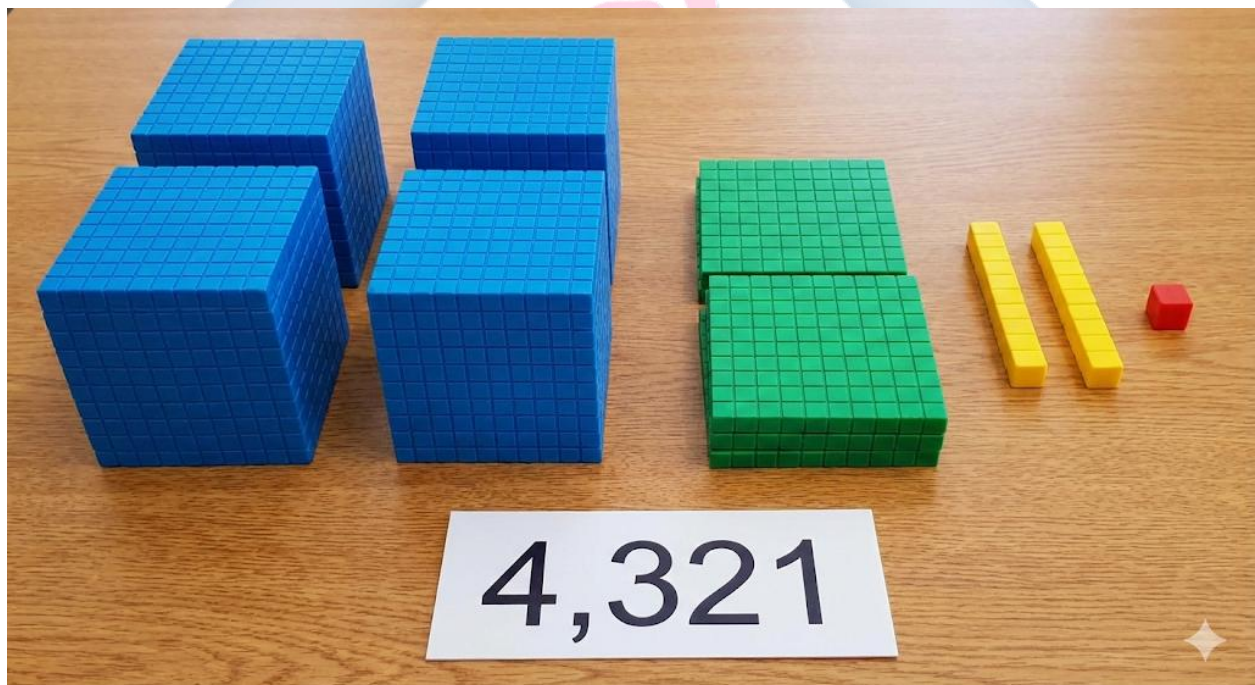
WEEK 1: WHOLE NUMBERS (PART 1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify the place value of digits in numbers up to 1,000 (Thousands, Hundreds, Tens, and Ones).
2. Read and write numbers in figures and words up to 1,000.
3. Expand whole numbers using their place values.
4. Compare and order numbers using the symbols $>$ (greater than), $<$ (less than), and $=$ (equal to).

CONTENT



Place Value (Thousands, Hundreds, Tens, Ones)

Think of a number like a house. Each digit lives in its own room. The room tells us its value. For numbers up to 1,000, we have four rooms: **Thousands (Th)**, **Hundreds (H)**, **Tens (T)**, and **Ones (O)**.

- **Example: 1,452**

- The digit **1** is in the **Thousands** room. Its value is **1000**.
- The digit **4** is in the **Hundreds** room. Its value is **400**.
- The digit **5** is in the **Tens** room. Its value is **50**.
- The digit **2** is in the **Ones** room. Its value is **2**.

Imagine you have 1,452 Naira. You have 1 one-thousand Naira note (₦1000), 4 five-hundred Naira notes (₦400), 5 ten Naira coins (₦50), and 2 one Naira coins (₦2).

Reading and Writing Numbers

Numbers can be written in **figures** (like 250) or **words** (like "two hundred and fifty").

- **Figures to Words:** 3,450 is "Three thousand, four hundred and fifty."
- **Words to Figures:** "Six hundred and five" is **605**.

Note: We use the word '**and**' to join the hundreds to the tens and ones. For example, 615 is "six hundred **and** fifteen."

Expanded Notation

Expanding a number means breaking it apart to show the full value of each digit. It's like unpacking a bag to see everything inside.

- **Example 1: Expand 738.**
 - 7 Hundreds + 3 Tens + 8 Ones.
 - **Expansion:** $700 + 30 + 8$.
- **Example 2: Expand 2,504.**
 - 2 Thousands + 5 Hundreds + 0 Tens + 4 Ones.
 - **Expansion:** $2000 + 500 + 0 + 4$. (We still write the 0 to show there are no tens).

Comparing and Ordering Numbers

We use special symbols to compare numbers, like using a see-saw. The bigger number is heavier and goes down.

- **> means Greater Than** (The open mouth eats the bigger number). Example: $10 > 6$.
- **< means Less Than** (The small point faces the smaller number). Example: $3 < 8$.
- **= means Equal To** (Both sides are the same). Example: $7 = 7$.

How to Compare:

1. Start comparing from the left-most digit (the biggest place value).
2. **Example:** Compare **540** and **520**.
 - Hundreds: 5 and 5 are the same.
 - Move to Tens: 4 and 2. 4 is bigger than 2.
 - So, **540 > 520**.

Ordering:

- **Ascending Order:** Arranging from the **smallest to the biggest** (like climbing stairs up). Example: 50, 100, 200, 500.
- **Descending Order:** Arranging from the **biggest to the smallest** (like walking down stairs). Example: 500, 200, 100, 50.

EVALUATION

1. What is the place value and value of the digit 6 in 6,782?
2. Write "Two thousand and fifteen" in figures.
3. Write 904 in words.
4. Expand the number 456 using place values (e.g., $400 + \dots$).
5. Use $>$, $<$, or $=$ to fill the gap: $800 \underline{\hspace{1cm}} 801$.
6. Use $>$, $<$, or $=$ to fill the gap: $150 \underline{\hspace{1cm}} 150$.
7. Arrange these numbers in ascending order: 200, 100, 500, 50.

ASSIGNMENT

1. **Oral/Written:** Read these numbers aloud to your parent, then write them in words: (a) 1,200 (b) 555 (c) 901.
2. **Written:** Expand these numbers: (a) 342 (b) 890.
3. **Practical:** Find three price tags at home or in the market (e.g., ₦250, ₦100, ₦700). Write them down and arrange them in descending order.



WEEK 2: WHOLE NUMBERS (PART 2)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Count in steps of 2s, 5s, 10s, and 100s to complete number sequences.
2. Identify odd and even numbers up to 1,000.
3. Round whole numbers to the nearest 10.
4. Round whole numbers to the nearest 100.

CONTENT



Skip Counting and Patterns

Skip counting is like jumping on a number line. It helps us count quickly. We jump by the same number each time.

- **Counting in 2s:** 2, 4, 6, 8, 10... (Like counting pairs of shoes).
- **Counting in 5s:** 5, 10, 15, 20... (Like counting fingers on hands).
- **Counting in 10s:** 10, 20, 30, 40... (Like counting bundles of ten sticks).

- **Counting in 100s:** 100, 200, 300, 400... (Like counting one-hundred Naira notes).

Odd and Even Numbers

- **Even Numbers:** Numbers that can be shared fairly between two people with nothing left over. They always end in **0, 2, 4, 6, or 8**.
 - *Examples:* 12 (can be shared as 6 and 6), 44, 100, 568.
- **Odd Numbers:** Numbers that cannot be shared fairly between two people; there is always one left over. They always end in **1, 3, 5, 7, or 9**.
 - *Examples:* 11 (shared as 5 and 5, with 1 left), 35, 79, 903.

Rounding Numbers to the Nearest 10

Rounding helps us find a friendly, close number. Think of the numbers on a street. Is a house number closer to 40 or 50?

- Look at the **digit in the Ones place**.
 - If it is **0, 1, 2, 3, or 4**, we round **DOWN** to the previous ten. The tens digit stays the same.
 - *Example:* 42 rounds down to 40. (42 is closer to 40 than to 50).
 - If it is **5, 6, 7, 8, or 9**, we round **UP** to the next ten. We add 1 to the tens digit.
 - *Example:* 47 rounds up to 50. (47 is closer to 50 than to 40).

Rounding Numbers to the Nearest 100

Now, think of a bigger street with house numbers like 300, 400, 500.

- Look at the **digit in the Tens place**.
 - If it is **0, 1, 2, 3, or 4**, we round **DOWN**. The hundreds digit stays the same.
 - *Example:* 320 rounds down to 300. (320 is closer to 300 than to 400).
 - If it is **5, 6, 7, 8, or 9**, we round **UP**. We add 1 to the hundreds digit.
 - *Example:* 380 rounds up to 400. (380 is closer to 400 than to 300).

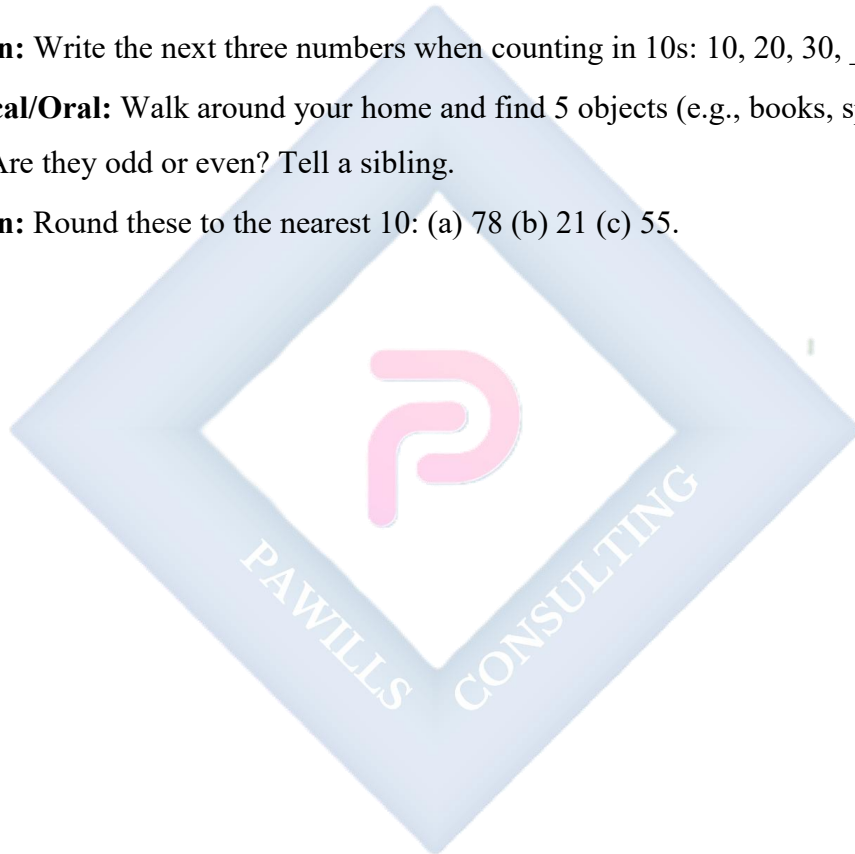
EVALUATION

1. Complete the pattern: 20, 22, 24, ____, ____.

2. Complete the pattern: 50, 100, 150, ___, ___.
3. Is 45 an Odd or Even number? How do you know?
4. Is 88 an Odd or Even number? How do you know?
5. Round 63 to the nearest 10.
6. Round 125 to the nearest 100.
7. Round 450 to the nearest 100.

ASSIGNMENT

1. **Written:** Write the next three numbers when counting in 10s: 10, 20, 30, ___, ___, ___.
2. **Practical/Oral:** Walk around your home and find 5 objects (e.g., books, spoons). Count them. Are they odd or even? Tell a sibling.
3. **Written:** Round these to the nearest 10: (a) 78 (b) 21 (c) 55.



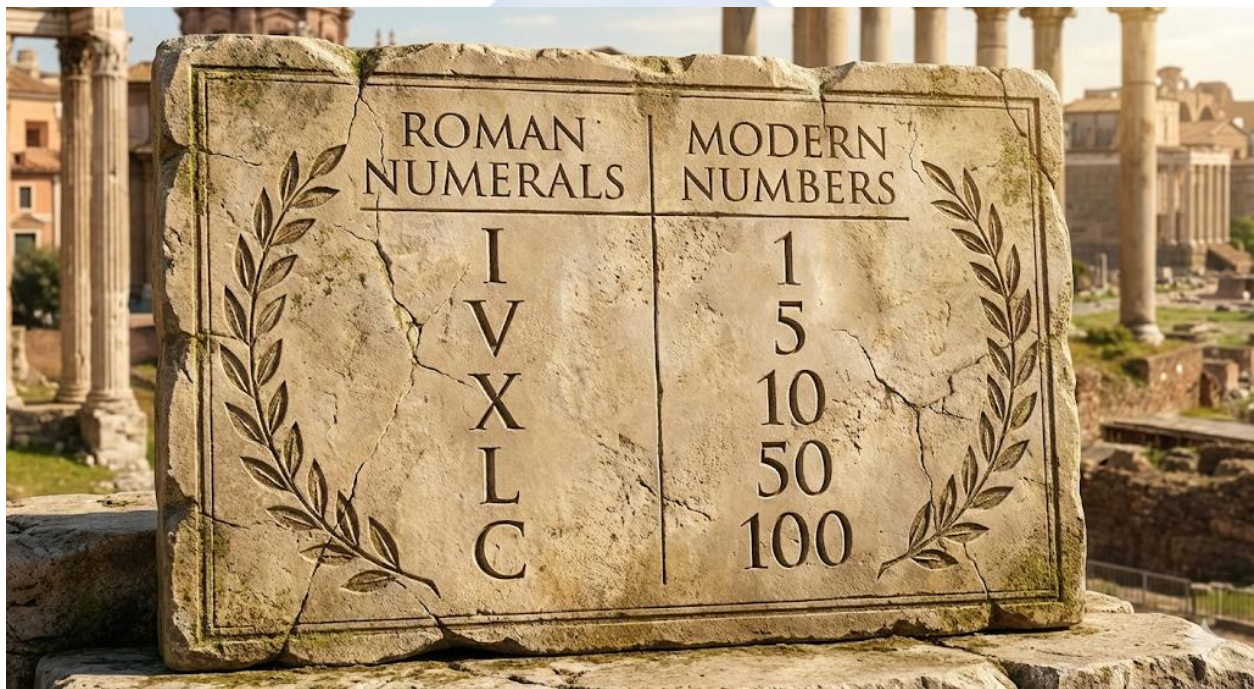
WEEK 3: WHOLE NUMBERS (PART 3)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify the basic Roman numerals (I, V, X, L, C).
2. Convert simple numbers from Hindu-Arabic (1, 2, 3) to Roman Numerals and vice versa.
3. List the first five prime numbers.
4. Identify square numbers up to 25.

CONTENT



Roman Numerals

Long ago, Romans used letters from their alphabet as numbers. We still see them on clocks, in book chapters, and for important dates.

Here are the basic letters and their values: I = 1 V = 5 X = 10 L = 50 C = 100

Rules for Reading Roman Numerals:

- **Addition Rule:** If a smaller letter is written **after** a bigger one, you **add** its value.

- *Example:* **VI** = V (5) + I (1) = **6**.
- *Example:* **XII** = X (10) + I (1) + I (1) = **12**.
- **Subtraction Rule:** If a smaller letter is written **before** a bigger one, you **subtract** its value.
 - *Example:* **IV** = V (5) - I (1) = **4**. (I before V)
 - *Example:* **IX** = X (10) - I (1) = **9**. (I before X)

Prime Numbers

A prime number is a special number. It has only **two factors**: 1 and itself. This means it can only be divided by 1 and itself without a remainder. You cannot make it by multiplying other whole numbers (except 1).

- **Examples:** 2, 3, 5, 7, 11.
- **Important:** **1 is not a prime number** because it has only one factor (itself).
- **2 is the only even prime number.** All other even numbers can be divided by 2.

Square Numbers

A square number is what you get when you multiply a whole number by itself. It's like finding the area of a square where all sides are equal.

- $1 \times 1 = 1$ (1 squared, like a tiny square)
- $2 \times 2 = 4$ (2 squared, imagine 2 rows of 2 dots = 4 dots)
- $3 \times 3 = 9$ (3 squared)
- $4 \times 4 = 16$ (4 squared)
- $5 \times 5 = 25$ (5 squared, up to here for Primary 4)

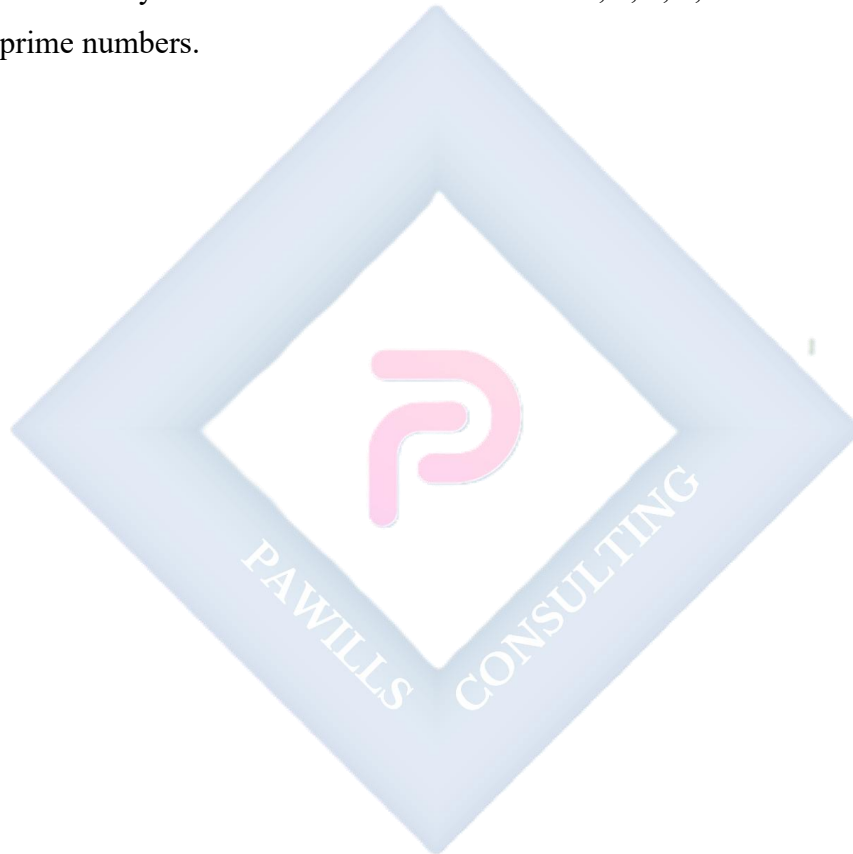
EVALUATION

1. Write the number 10 in Roman Numerals.
2. What Hindu-Arabic number is **XV**?
3. Write the number 4 in Roman Numerals.
4. List all the prime numbers between 1 and 10.

5. What is 3 squared? (What is 3×3 ?)
6. Is 9 a prime number? Why or why not?
7. Write 20 in Roman Numerals.

ASSIGNMENT

1. **Written:** Convert these to Roman Numerals: (a) 6 (b) 15 (c) 50.
2. **Drawing/Practical:** Draw dots to show these square numbers: 4 (2×2) and 9 (3×3).
3. **Oral/Written:** Say these numbers aloud to a friend: 2, 4, 5, 8, 11. Ask them to clap only for the prime numbers.



WEEK 4: FRACTIONS (PART 1)

OBJECTIVES

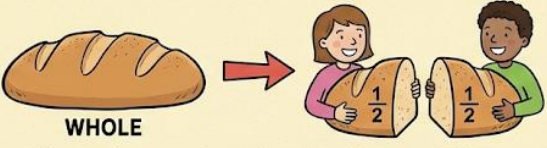
By the end of the lesson, pupils should be able to:

1. Define a fraction as a part of a whole.
2. Identify the numerator and denominator in a fraction.
3. Recognize equivalent fractions using diagrams.
4. Compare simple fractions with the same denominator.

CONTENT

UNDERSTANDING FRACTIONS

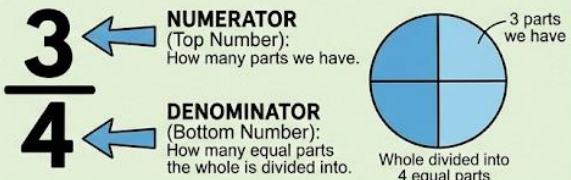
WHAT IS A FRACTION?



WHOLE

A fraction is an equal part of a whole thing or a whole group. Sharing a loaf equally gives each a "one-half" ($\frac{1}{2}$).

PARTS OF A FRACTION



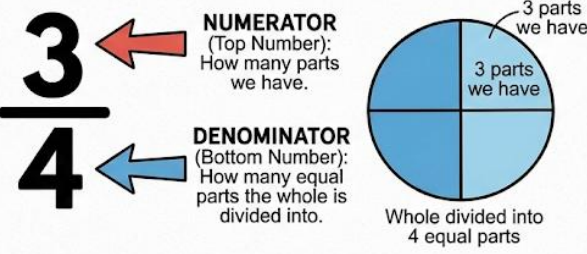
NUMERATOR
(Top Number):
How many parts we have.

DENOMINATOR
(Bottom Number):
How many equal parts the whole is divided into.

3 parts we have

Whole divided into 4 equal parts

PARTS OF A FRACTION



NUMERATOR
(Top Number):
How many parts we have.

DENOMINATOR
(Bottom Number):
How many equal parts the whole is divided into.

3 parts we have

Whole divided into 4 equal parts

EQUIVALENT FRACTIONS



$\frac{1}{2}$ $\frac{2}{4}$ $\frac{3}{6}$

Different fractions, same value!
The shaded area is the same amount. So, $\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$.

COMPARING FRACTIONS (Same Denominator)



$\frac{1}{5}$ $\frac{3}{5}$

Same-sized pieces (fifths). Just compare the numerators!
3 pieces are more than 1 piece. So, $\frac{3}{5} > \frac{1}{5}$.

Understanding Fractions

A fraction is an equal part of a whole thing or a whole group. If you share a loaf of bread equally with your friend, you each get a fraction called **one-half** ($\frac{1}{2}$).

Parts of a Fraction

A fraction is written as two numbers with a line between them.

- **Top Number (Numerator):** This tells **how many parts** we are talking about or have. It *numbers* the parts.
- **Bottom Number (Denominator):** This tells **how many equal parts** the whole is divided into. It *names* the type of fraction (halves, thirds, quarters).

Example: In the fraction $\frac{3}{4}$ (three-quarters):

- **3** is the **Numerator** (we have 3 parts).
- **4** is the **Denominator** (the whole is cut into 4 equal parts).

Equivalent Fractions

These are different fractions that have the **same value** or represent the same amount. Think of cutting the same cake differently.

- If you have $\frac{1}{2}$ of a cake, it is the same amount as having $\frac{2}{4}$ of the same cake.
- Look at this diagram:

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$$

[##] [##] [###]

[] [##] [###]

The shaded area is the same! So, $\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$.

Comparing Fractions (Same Denominator)

This is easy! If the denominators (bottom numbers) are the same, it means the wholes are cut into the same-sized pieces. You just compare the numerators (top numbers).

- *Example:* Compare $\frac{1}{5}$ and $\frac{3}{5}$.
 - Both wholes are cut into **5** equal pieces (fifths).
 - Which is more: **1 piece** or **3 pieces**? 3 pieces is more.
 - So, $\frac{3}{5} > \frac{1}{5}$.

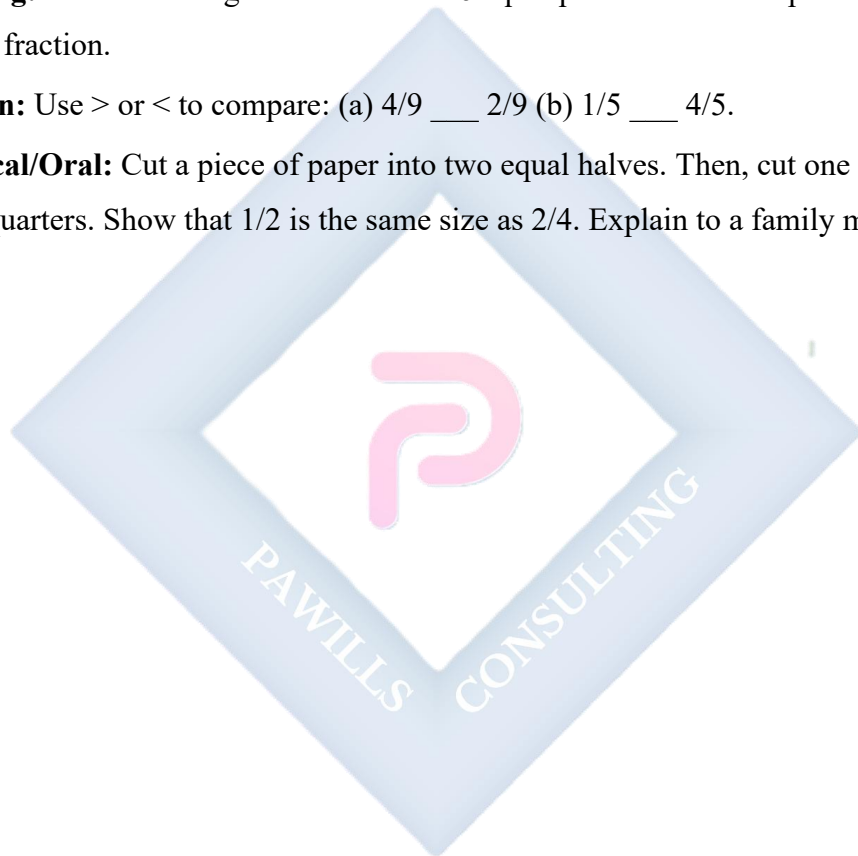
EVALUATION

1. In the fraction $\frac{2}{3}$, which number is the denominator?

2. Draw a circle, divide it into 4 equal parts, and shade 1 part. What fraction is shaded?
3. Which fraction is bigger: $\frac{2}{7}$ or $\frac{5}{7}$?
4. Write two fractions that are equivalent to $\frac{1}{2}$.
5. If I eat $\frac{1}{8}$ of a cake and you eat $\frac{3}{8}$ of the same cake, who ate more?
6. What does the denominator of a fraction tell us?

ASSIGNMENT

1. **Drawing:** Draw a rectangle. Divide it into 6 equal parts and shade 2 parts. Write the shaded fraction.
2. **Written:** Use $>$ or $<$ to compare: (a) $\frac{4}{9}$ ____ $\frac{2}{9}$ (b) $\frac{1}{5}$ ____ $\frac{4}{5}$.
3. **Practical/Oral:** Cut a piece of paper into two equal halves. Then, cut one half into two equal quarters. Show that $\frac{1}{2}$ is the same size as $\frac{2}{4}$. Explain to a family member.



WEEK 5: MID-TERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Attempt questions covering Weeks 1–4 independently.

CONTENT

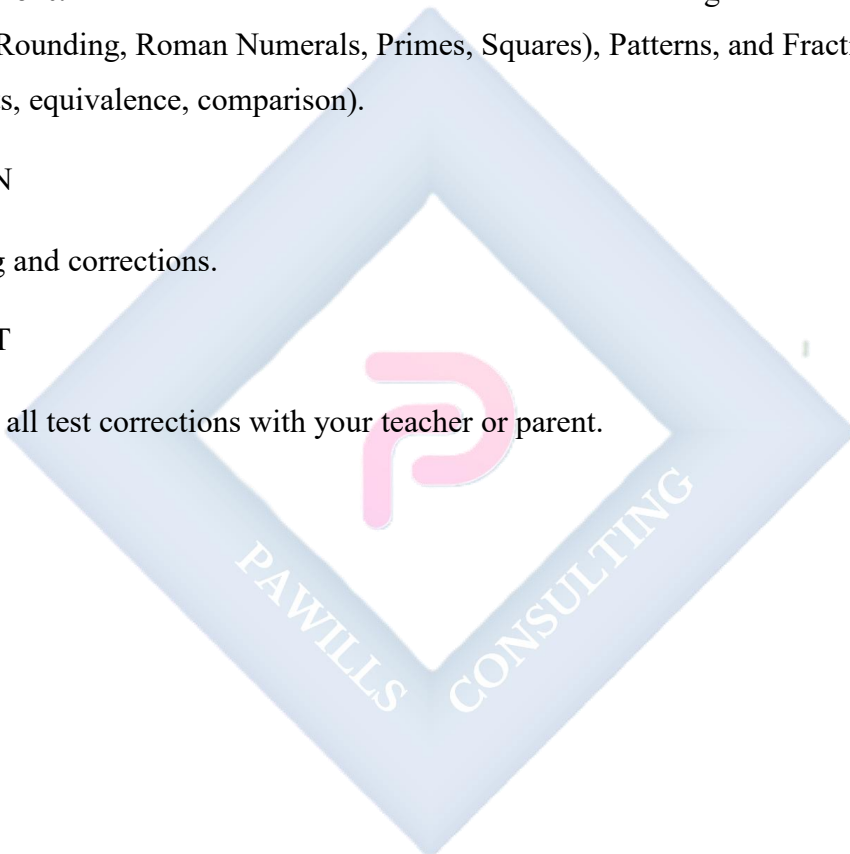
- **Assessment:** The teacher will administer a written test covering Whole Numbers (Place Value, Rounding, Roman Numerals, Primes, Squares), Patterns, and Fractions (Basic concepts, equivalence, comparison).

EVALUATION

- Grading and corrections.

ASSIGNMENT

- Review all test corrections with your teacher or parent.



WEEK 6: MID-TERM BREAK



WEEK 7: ADDITION AND SUBTRACTION (PART 1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Add 2-digit and 3-digit numbers without regrouping.
2. Add numbers with regrouping (carrying).
3. Solve simple word problems involving addition.

CONTENT

ADDITION WITHOUT REGROUPING

Step 1: $3+5=8$ (Ones)
Step 2: $2+4=6$ (Tens)
Step 3: $1+0=1$ (Hundreds)

Simple addition where the sum in each column is less than 10. Always start from the Ones column.

$10 + 0 = 10$

ADDITION WITH REGROUPING (Carrying)

(Carry 1)
Ones: $8+7=15$. Write 5 in Ones column, carry 1 to Tens column.
Tens: 1 (carried) + $5+2=8$. Write 8.
Hundreds: $3+0=3$. Write 3.

Happens when the sum in a column is 10 or more. Write ones digit, carry tens digit.

$10 + 10 = 20$

WORD PROBLEMS (Addition)

Look for keywords: total, sum, altogether, in all, combined.

Example: Musa has 250 yams. Ada gives him 150 more yams. How many yams does he have altogether?

Think: Start with 250, add 150 more.

Musa: 250 yams
Ada: 150 yams

Calculation: $250 + 150 = 400$

Answer: He has 400 yams altogether.

Total: 400 yams

Addition Without Regrouping

This is simple addition where the sum in each column is less than 10. We always start from the **Ones column (right side)**, then move left to Tens, then Hundreds.

- **Example:** $123 + 45$.

$$\begin{array}{r} \text{H T O} \\ 123 \\ + 45 \\ \hline \end{array}$$

$$\begin{array}{r} \text{-----} \\ 1\ 6\ 8 \end{array}$$

Steps: $3+5=8$ (Ones), $2+4=6$ (Tens), $1+0=1$ (Hundreds).

Addition With Regrouping (Carrying)

This happens when the sum in a column is **10 or more**. We write the ones digit of the answer in that column and **carry** the tens digit over to the next column on the left.

- **Example:** $358 + 27$.

$$\begin{array}{r} \text{(Carry 1)} \\ \text{H T O} \\ 3\ 5\ 8 \\ +\ 2\ 7 \\ \text{-----} \\ 3\ 8\ 5 \end{array}$$

Steps:

- Ones:** $8 + 7 = 15$. Write **5** in Ones column, **carry 1** to Tens column.
- Tens:** 1 (carried) $+ 5 + 2 = 8$. Write **8**.
- Hundreds:** $3 + 0 = 3$. Write **3**.

Word Problems (Addition)

Look for keywords that tell you to add: "**total**," "**sum**," "**altogether**," "**in all**," "**combined**."

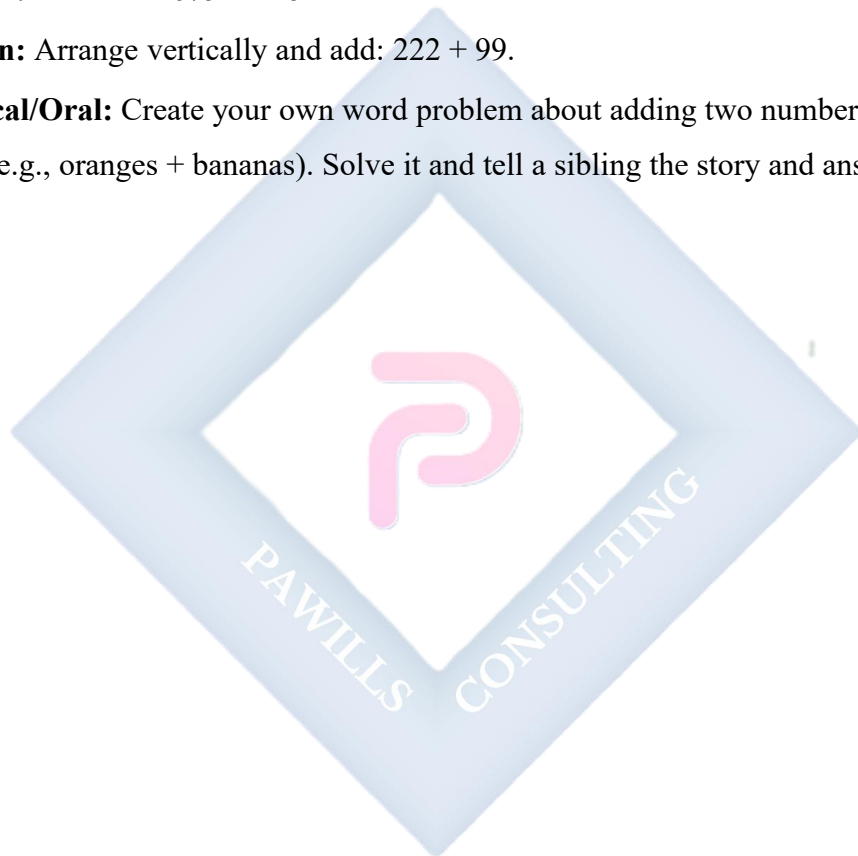
- *Example:* Musa has 250 yams. Ada gives him 150 more yams. How many yams does he have **altogether**?
 - Think: Start with 250, add 150 more.
 - Calculation: $250 + 150 = 400$.
 - Answer: He has **400** yams altogether.

EVALUATION

1. Add: $340 + 20$.
2. Add: $56 + 13$.
3. Solve: $457 + 38$ (Show your working with carrying).
4. Find the sum of 120, 30, and 5.
5. A boy has 15 marbles and buys 16 more. How many marbles does he have in total?

ASSIGNMENT

1. **Written:** Calculate: $675 + 248$.
2. **Written:** Arrange vertically and add: $222 + 99$.
3. **Practical/Oral:** Create your own word problem about adding two numbers of items at home (e.g., oranges + bananas). Solve it and tell a sibling the story and answer.





WEEK 8: ADDITION AND SUBTRACTION (PART 2)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Subtract 2-digit and 3-digit numbers without regrouping.
2. Subtract numbers with regrouping (borrowing).
3. Solve subtraction problems involving zeros.
4. Solve simple word problems involving subtraction.

CONTENT

SUBTRACTION METHODS & WORD PROBLEMS

SUBTRACTION WITHOUT REGROUPING

This is simple subtraction where each digit on top is larger than the digit below it. Start from the Ones column.

$$\begin{array}{r} \text{T} \quad \text{O} \\ 68 \\ - 42 \\ \hline 26 \end{array}$$

Steps:
8 - 2 = 6 (Ones),
6 - 4 = 2 (Tens).

SUBTRACTION WITH REGROUPING (BORROWING)

This happens when a digit in the top number is smaller than the digit below it. We must borrow 1 from the next column on the left. This '1' is worth 10 in the current column.

$$\begin{array}{r} \text{T} \quad \text{O} \\ 52 \\ - 38 \\ \hline 14 \end{array}$$

Steps:
a. Ones: 2 is smaller than 8. We can't subtract. Borrow 1 from the Tens column. The '5' becomes '4', and the '2' becomes '12'.
b. Now subtract Ones: 12 - 8 = 4.
c. Tens: 4 (left after borrowing) - 3 = 1.

SUBTRACTION ACROSS ZEROS

This is tricky because you might have to borrow from several columns. Remember, you can't borrow from zero; you go to the next non-zero digit.

$$\begin{array}{r} \text{H} \quad \text{T} \quad \text{O} \\ 500 \\ - 237 \\ \hline 263 \end{array}$$

Steps:
a. Ones: 0 - 7? Can't. Need to borrow. Tens column is 0, can't borrow. Go to Hundreds.
b. Borrow 1 from 5 (Hundreds). 5 becomes 4. The borrowed 1 (100) goes to Tens as 10 tens.
c. Now, borrow 1 ten from the Tens (now has 10). Tens become 9. The borrowed 1 ten (10) goes to Ones as 10 ones.
d. Now subtract:
Ones: 10 - 7 = 3.
Tens: 9 - 3 = 6.
Hundreds: 4 - 2 = 2.

WORD PROBLEMS (SUBTRACTION)

Look for keywords: "difference," "remain," "left," "take away," "how much less."

Look for keywords: "difference," "left," "take away," "how many more," "how much less."

Example: There are 50 pupils in a class. 20 are girls. How many are boys?

Think: Total pupils minus girls equals boys.
Calculation: 50 - 20 = 30.
Answer: There are 30 boys.

Subtraction Without Regrouping

This is simple subtraction where each digit on top is larger than the digit below it. Start from the Ones column.

- Example: 68 - 42.

T O

6 8

$$\begin{array}{r} -42 \\ \hline \end{array}$$

$$26$$

Steps: $8 - 2 = 6$ (Ones), $6 - 4 = 2$ (Tens).

Subtraction With Regrouping (Borrowing)

This happens when a digit in the top number is **smaller** than the digit below it. We must **borrow** **1** from the next column on the left. This '1' is worth 10 in the current column.

- **Example:** $52 - 38$.

T O

$$52$$

$$\begin{array}{r} -38 \\ \hline \end{array}$$

$$14$$

Steps:

- Ones:** 2 is smaller than 8. We can't subtract. Borrow 1 from the Tens column. The '5' becomes '4', and the '2' becomes '12'.
- Now subtract Ones: $12 - 8 = 4$.
- Tens:** 4 (left after borrowing) - 3 = 1.

Subtraction Across Zeros

This is tricky because you might have to borrow from several columns. Remember, you can't borrow from zero; you go to the next non-zero digit.

- **Example:** $500 - 237$.

H T O

$$500$$

$$\begin{array}{r} -237 \\ \hline \end{array}$$

2 6 3

Steps:

- a. **Ones:** $0 - 7$? Can't. Need to borrow. Tens column is 0, can't borrow. Go to Hundreds.
- b. Borrow 1 from 5 (Hundreds). 5 becomes 4. The borrowed 1 (100) goes to Tens as 10 tens.
- c. Now, borrow 1 ten from the Tens (now has 10). Tens become 9. The borrowed 1 ten (10) goes to Ones as 10 ones.
- d. Now subtract: Ones: $10 - 7 = 3$. Tens: $9 - 3 = 6$. Hundreds: $4 - 2 = 2$.

Word Problems (Subtraction)

Look for keywords: "difference," "remain," "left," "take away," "how many more," "how much less."

- *Example:* There are 50 pupils in a class. 20 are girls. How many are boys?
 - Think: Total pupils minus girls equals boys.
 - Calculation: $50 - 20 = 30$.
 - Answer: There are **30** boys.

EVALUATION

1. Subtract: $99 - 45$.
2. Solve: $71 - 35$ (Show borrowing).
3. Find the difference between 500 and 120.
4. Solve: $403 - 186$ (Be careful with borrowing).
5. Bola had 50 biscuits. She gave 20 to her friends. How many biscuits does she have left?

ASSIGNMENT

1. **Written:** Calculate: $800 - 455$.
2. **Written:** Solve: $92 - 68$.

3. **Practical/Oral:** With a parent, look at a price tag for an item (e.g., ₦500). Ask: "If I pay with ₦1000, how much change should I get?" Calculate the subtraction.



WEEK 9: ADDITION AND SUBTRACTION (PART 3)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Solve mixed addition and subtraction problems.
2. Add and subtract money (Naira and Kobo).
3. Check subtraction answers using addition.
4. Apply strategies to real-life shopping scenarios.

CONTENT

MONEY: NAIRA (₦) & KOBO (k)

Our money is in Naira (₦) and Kobo (k).
100 Kobo = 1 Naira.
When writing with decimals, ₦2.50 means
2 Naira and 50 Kobo.



EXAMPLE: ADD ₦2.50 AND ₦1.50

₦	k
2	50
+	150
4	00

Steps
50k + 50k = 100k = ₦1.
Write 0 in kobo column,
carry ₦1 to Naira column.
₦2 + ₦1 + ₦1 (carried) = ₦4.



CHECKING SUBTRACTION WITH ADDITION

This is a smart way to see if your subtraction is correct. It uses the fact that addition and subtraction are opposites (inverse operations).



TO CHECK IF $10 - 3 = 7$ IS CORRECT:

Step 1: Add the answer (7) to the smaller number (3).
Step 2: If the sum equals the original big number (10), then the subtraction is correct.
Step 3: $7 + 3 = 10$. ✓ Correct.

REAL-LIFE SHOPPING (MIXED OPERATIONS)

Often, we add to find a total cost, then subtract from our money to find change.

SCENARIO: You have ₦500. You buy bread for ₦200 and milk for ₦150.



a. Add to find total spent:
 $₦200 + ₦150 = ₦350$.

b. Subtract total spent from your money to find change:
 $₦500 - ₦350 = ₦150$.

ANSWER: You will get ₦150 change.

EVALUATION

- 1 Add: $₦120 + ₦80$.
- 2 Subtract: $₦200 - ₦50$.
- 3 Solve: $15 + 10 - 5$.
- 4 Check this subtraction using addition: $50 - 20 = 30$.
- 5 If a book costs ₦100 and a pen costs ₦50, what is the total cost? If you pay with ₦200, how much change do you get?

Money: Naira and Kobo

Our money is in Naira (₦) and Kobo (k). **100 Kobo = 1 Naira.** When writing with decimals, ₦2.50 means 2 Naira and 50 Kobo. Always line up the decimal points when adding or subtracting.

- **Example:** Add ₦2.50 and ₦1.50.

$$\begin{array}{r}
 \text{₦ } k \\
 2.50 \\
 + 1.50 \\
 \hline
 4.00
 \end{array}$$

Steps: $50k + 50k = 100k = \text{₦}1$. Write 0 in kobo column, carry ₦1 to Naira column. $\text{₦}2 + \text{₦}1 + \text{₦}1$ (carried) = ₦4.

Checking Subtraction with Addition

This is a smart way to see if your subtraction is correct. It uses the fact that addition and subtraction are opposites (inverse operations).

- To check if $10 - 3 = 7$ is correct:
 - Add the **answer (7)** to the **smaller number (3)**.
 - If the sum equals the **original big number (10)**, then the subtraction is correct.
 - $7 + 3 = 10$. ✓ Correct.

Real-life Shopping (Mixed Operations)

Often, we add to find a total cost, then subtract from our money to find change.

- *Scenario:* You have ₦500. You buy bread for ₦200 and milk for ₦150.
 - a. **Add** to find total spent: $\text{₦}200 + \text{₦}150 = \text{₦}350$.
 - b. **Subtract** total spent from your money to find change: $\text{₦}500 - \text{₦}350 = \text{₦}150$.
 - Answer: You will get **₦150** change.

EVALUATION

1. Add: $\text{₦}120 + \text{₦}80$.
2. Subtract: $\text{₦}200 - \text{₦}50$.
3. Solve: $15 + 10 - 5$.
4. Check this subtraction using addition: $50 - 20 = 30$.

5. If a book costs ₦100 and a pen costs ₦50, what is the total cost? If you pay with ₦200, how much change do you get?

ASSIGNMENT

1. **Written:** Calculate: ₦1000 - ₦650.
2. **Written:** Find the sum of ₦45.50 and ₦20.50.
3. **Practical/Oral:** With a family member, pretend to buy two items from your cupboard. Decide on prices (under ₦500). Calculate the total cost and the change from ₦500. Explain your steps.



WEEK 10: MULTIPLICATION (PART 1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define multiplication as repeated addition.
2. Recite multiplication tables 2, 3, 4, 5, and 10.
3. Demonstrate the commutative property of multiplication.
4. Multiply 2-digit numbers by 1-digit numbers (no regrouping).

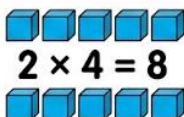
CONTENT

UNDERSTANDING MULTIPLICATION

MULTIPLICATION AS REPEATED ADDITION


 $2 + 2 + 2 = 6$
3 groups of 2 = 6
 $3 \times 2 = 6$ ← means "groups of" or "times"

COMMUTATIVE PROPERTY


 $2 \times 4 = 8$

 $4 \times 2 = 8$
Swapping numbers, same answer:
 $2 \times 4 = 4 \times 2$

MULTIPLYING BY 10 & 100

$5 \times 10 = 50$ ← To multiply by 10, add one zero.
 $5 \times 100 = 500$ ← To multiply by 100, add two zeros.

MULTIPLYING 2-DIGIT BY 1-DIGIT

$\begin{array}{r} \text{T} \quad \text{O} \\ 23 \\ \times 2 \\ \hline 46 \end{array}$ a. Ones: $3 \times 2 = 6$ (Write 6)
b. Tens: $2 \times 2 = 4$ (Write 4)

EVALUATION QUESTIONS

1. Write $4 + 4 + 4$ as multiplication.
2. What is 5×3 ?
3. Solve: 12×10 .
4. Complete: $4 \times 2 = \underline{\quad} \times 4$.
5. Multiply: 32×3 .
6. One tricycle has 3 wheels. How many wheels do 5 tricycles have?

Multiplication as Repeated Addition

Multiplication is a quick way to add the **same number** many times.

- *Example:* $2 + 2 + 2 = 6$.
 - This is **3 groups of 2**.
 - We write it as: $3 \times 2 = 6$.
 - The $\times$ sign means "groups of" or "times."

Commutative Property of Multiplication

This is a fancy name for a simple idea: **Swapping the numbers does not change the answer.**

- $3 \times 4 = 12$.
- $4 \times 3 = 12$.
- So, $3 \times 4 = 4 \times 3$. It works every time!

Multiplication by 10 and 100

These are special and easy rules:

- **To multiply by 10**, add **one zero** to the end of the number.
 - $5 \times 10 = 50$. $23 \times 10 = 230$.
- **To multiply by 100**, add **two zeros** to the end of the number.
 - $5 \times 100 = 500$. $7 \times 100 = 700$.

Multiplying 2-Digit by 1-Digit (No Regrouping)

We multiply each digit of the 2-digit number by the 1-digit number, starting from the **Ones place**.

- **Example:** 23×2 .

$$\begin{array}{r} \text{T O} \\ 23 \\ \times 2 \\ \hline 46 \end{array}$$

Steps:

- a. Multiply Ones: $3 \times 2 = 6$. Write 6 in Ones place.
- b. Multiply Tens: $2 \times 2 = 4$. Write 4 in Tens place.

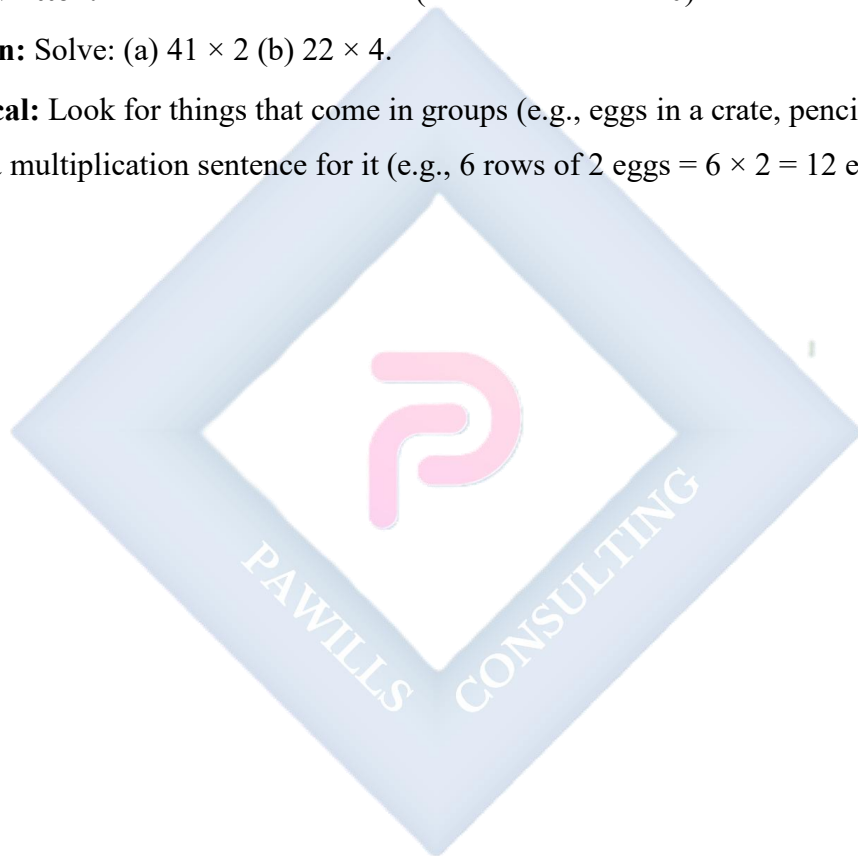
EVALUATION

1. Write $4 + 4 + 4$ as a multiplication sentence.

2. What is 5×3 ?
3. Solve: 12×10 .
4. Complete: $4 \times 2 = \underline{\quad} \times 4$.
5. Multiply: 32×3 .
6. One tricycle has 3 wheels. How many wheels do 5 tricycles have?

ASSIGNMENT

1. **Oral/Written:** Recite the 4 times table (from 4×1 to 4×10) to someone at home.
2. **Written:** Solve: (a) 41×2 (b) 22×4 .
3. **Practical:** Look for things that come in groups (e.g., eggs in a crate, pencils in a pack).
Write a multiplication sentence for it (e.g., 6 rows of 2 eggs = $6 \times 2 = 12$ eggs).





WEEK 11:REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts covering Whole Numbers, Fractions, Addition, Subtraction, and Multiplication.
2. Attempt practice questions confidently.

CONTENT

- **Review – Whole Numbers:** Place value (Th, H, T, O). Rounding to nearest 10/100. Odd/Even. Prime numbers (2,3,5,7,11). Square numbers (1,4,9,16,25).
- **Review – Fractions:** Meaning, numerator/denominator. Comparing fractions with same denominator (bigger numerator = bigger fraction).
- **Review – Addition & Subtraction:** Vertical method. Key words: "sum/altogether" for addition, "difference/left" for subtraction. Carrying and borrowing.
- **Review – Multiplication:** Repeated addition. Times tables (2,3,4,5,10). Multiplying by 10/100. Simple 2-digit $\times$ 1-digit.

EVALUATION

- Mixed practice quiz covering all term's work (e.g., 5 questions from different topics).

ASSIGNMENT

- Study all notes for the end-of-term examination. Practice 5 addition, 5 subtraction, and 5 multiplication problems from your textbook.

PRIMARY 4 MATHEMATICS SCHEME OF WORK

SECOND TERM

Week	Topic	Breakdown (Subtopics)
1	Multiplication (Part 2)	• Multiplication tables: 6, 7, 8, 9, 11, 12 times tables • Multiplication of 2-digit by 1-digit numbers (with regrouping) • Multiplication of 2-digit by 2-digit numbers (introduction) • Multiplication using grid method • Multiplication word problems • Real-life applications of multiplication
2	Division (Part 1)	• Understanding division as sharing and grouping • Division as inverse of multiplication • Division facts from multiplication tables • Division of 2-digit numbers by 1-digit numbers (no remainder) • Division with remainders • Recording division: $15 \div 3 = 5$ and $3 \overline{)15}$
3	Division (Part 2)	• Long division method (introduction) • Division of 3-digit numbers by 1-digit numbers • Checking division using multiplication • Division word problems • Real-life applications: sharing equally, grouping • Mixed operations problems
4	Estimation	• Understanding estimation: finding approximate answers • Rounding to nearest 10, 100, 1000 • Estimating sums and differences

		• Estimating products and quotients • When to use estimation (checking reasonableness of answers) • Real-life estimation situations
5	MID-TERM EXAMINATION	• Assessment of Weeks 1-4
6	MID-TERM BREAK	• Holiday/Rest
7	Open Sentences	• Understanding open sentences (equations with missing numbers) • Finding missing numbers in addition: $__ + 5 = 12$ • Finding missing numbers in subtraction: $15 - __ = 8$ • Finding missing numbers in multiplication: $4 \times __ = 20$ • Finding missing numbers in division: $__ \div 3 = 7$ • Using trial and error and inverse operations • Creating and solving open sentences
8	Money (Part 1)	• Identifying Nigerian currency: coins and notes • Naira and Kobo: ₦1 = 100k • Converting between Naira and Kobo • Counting money: notes and coins combinations • Addition and subtraction of money • Making change from purchases • Word problems involving money
9	Money (Part 2)	• Multiplication and division of money • Shopping calculations: total cost, change • Budgeting: planning how to spend money • Profit and loss (simple introduction) • Saving money: simple interest concept (awareness)

		• Real-life money problems and scenarios
10	Length	• Understanding length: how long something is • Units of length: millimeter (mm), centimeter (cm), meter (m), kilometer (km) • Conversions: $10\text{mm} = 1\text{cm}$, $100\text{cm} = 1\text{m}$, $1000\text{m} = 1\text{km}$ • Measuring length using rulers and tape measures • Estimating lengths • Addition and subtraction of lengths • Perimeter: measuring around shapes • Word problems on length
11	REVISION	• Review: Multiplication, division, estimation, open sentences, money, length
12	EXAMINATION	• End of Term Assessment
13	CLOSING/VACATION	• Term ends

PRIMARY 4 MATHEMATICS LESSON NOTES

SECOND TERM

WEEK 1: MULTIPLICATION (PART 2)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recite the multiplication tables 6, 7, 8, 9, 11, and 12.
2. Multiply a 2-digit number by a 1-digit number with regrouping (carrying).
3. Solve multiplication of 2-digit by 2-digit numbers using the grid method.
4. Solve simple word problems involving multiplication in real-life situations.

CONTENT



MASTERING MULTIPLICATION



MULTIPLICATION TABLES - SUPERPOWER MATH!

Learning tables by heart helps solve problems fast! Practice with claps or rhymes.

6 TIMES TABLE (Insects)



6, 12, 18, 24, 30, 36, 42, 48, 54, 60

7 TIMES TABLE (Days in a Week)



7, 14, 21, 28, 35, 42, 49, 56, 63, 70

8 TIMES TABLE (Octopus Legs)



8, 16, 24, 32, 40, 48, 56, 64, 72, 80

9 TIMES TABLE (Sum of Digits = 9)



$1+8=9$, $2+7=9$
9, 18, 27, 36, 45, 54, 63, 72, 81, 90

11 TIMES TABLE (Double Digit up to 9)



11, 22, 33, 44, 55, 66, 77, 88, 99, 110

12 TIMES TABLE (Dozens)



12, 24, 36, 48, 60, 72, 84, 96, 108, 120

MULTIPLYING 2-DIGIT BY 1-DIGIT (WITH REGROUPING)

Like addition with carrying. Multiply Ones, write ones digit, carry tens digit.

Example: 26×4

$$\begin{array}{r} \text{T} \text{ O} \\ 26 \\ \times 4 \\ \hline 104 \end{array}$$

(Carry 2)

a. Multiply Ones: $4 \times 6 = 24$. Write 4 in Ones place. Carry 2 to Tens place.

b. Multiply Tens: $4 \times 2 = 8$. Add carried 2: $8 + 2 = 10$. Write 10.

c. Answer: 104.

MULTIPLYING 2-DIGIT BY 2-DIGIT (GRID METHOD)

Breaks numbers into Tens and Ones. Like a multiplication table for parts.

Example: 12×13

Break: $12 = 10 + 2$. $13 = 10 + 3$.

Grid:

	10	2
10	100	20
3	30	6

c. Add all answers:
 $100 + 20 + 30 + 6 = 156$.

WORD PROBLEMS - LOOK FOR CLUES!

Clues: 'product,' 'times,' 'each,' 'per,' 'total,' 'altogether' often mean multiply.

Example:
If one exercise book costs N50, how much will 5 books cost?



Think: Cost of one $\times$ Number of books.
Calculation: $N50 \times 5 = N250$.

Multiplication Tables

Learning tables by heart is like having a superpower for math. It helps you solve problems very fast. Let's practice the new tables together with claps or rhymes.

- **6 times table:** Think of insects with 6 legs! 6, 12, 18, 24, 30, 36, 42, 48, 54, 60.
- **7 times table:** Think of days in a week! 7, 14, 21, 28, 35, 42, 49, 56, 63, 70.
- **8 times table:** Think of octopus legs! 8, 16, 24, 32, 40, 48, 56, 64, 72, 80.
- **9 times table:** A fun trick: the digits of the answers add up to 9! 9, 18, 27, 36, 45, 54, 63, 72, 81, 90. ($1+8=9$, $2+7=9$).
- **11 times table:** Easy! Up to 9, it's just double the digit. 11, 22, 33, 44, 55, 66, 77, 88, 99, 110.
- **12 times table:** Think of dozens, like eggs. 12, 24, 36, 48, 60, 72, 84, 96, 108, 120.

Multiplying 2-Digit by 1-Digit (With Regrouping/Carrying)

This is like addition with carrying. When you multiply the Ones and get 10 or more, you write the ones digit and carry the tens digit to the Tens column.

- **Example:** 26×4

$$\begin{array}{r}
 \text{(Carry 2)} \\
 \text{T O} \\
 26 \\
 \times 4 \\
 \hline
 104
 \end{array}$$

Steps:

- Multiply Ones:** $4 \times 6 = 24$. Write **4** in the Ones place. **Carry 2** to the Tens place.
- Multiply Tens:** $4 \times 2 = 8$. Add the carried 2: $8 + 2 = 10$. Write 10 in the Tens and Hundreds place.
- Answer: 104.**

Multiplying 2-Digit by 2-Digit (Grid Method)

The grid method breaks numbers into Tens and Ones, which makes big multiplication easier to see. It is like a multiplication table for parts of the numbers.

- **Example:** 12×13

a. **Break the numbers:** $12 = 10 + 2$. $13 = 10 + 3$.

b. **Draw a grid** and multiply the parts:

$\times$	10	2
10	100	20
3	30	6

c. **Add all the answers inside the grid:** $100 + 20 + 30 + 6 = 156$.

Word Problems

Look for clues: "**product,**" "**times,**" "**each,**" "**per,**" "**total,**" "**altogether**" often mean multiply.

- *Example:* If one exercise book costs ₦50, how much will 5 books cost?
 - Think: **Cost of one $\times$ Number of books.**
 - Calculation: $\text{₦}50 \times 5 = \text{₦}250$.

EVALUATION

1. What is 6×7 ?
2. Solve: 45×3 (Show your working with carrying).
3. Use the grid method to solve 14×11 .
4. If a pack contains 12 pencils, how many pencils are in 5 packs?
5. What is 8×9 ?

ASSIGNMENT

1. **Oral:** Recite the 9 times table to a family member.
2. **Written:** Calculate: 38×5 . Show your carrying.
3. **Practical:** Look at a carton of eggs at home. If one row has a certain number of eggs (e.g., 6), count the rows. Write a multiplication sentence to find the total eggs.

WEEK 2: DIVISION (PART 1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define division as sharing equally or grouping.
2. Relate division to multiplication (Inverse operation).
3. Divide 2-digit numbers by 1-digit numbers without a remainder.
4. Divide numbers with a remainder.

CONTENT

UNDERSTANDING DIVISION

UNDERSTANDING DIVISION: SHARING & GROUPING

1. Sharing



How many does each person get?
(e.g., Share 10 sweets between 2 friends: $10 \div 2 = 5$ each).

2. Grouping



How many groups can we make?
(e.g., How many groups of 3 are in 12? $12 \div 3 = 4$ groups).

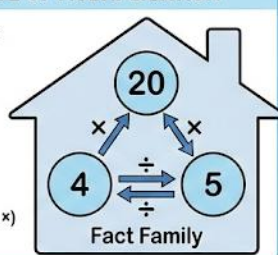
DIVISION AS INVERSE OF MULTIPLICATION

Division is the exact opposite of multiplication.

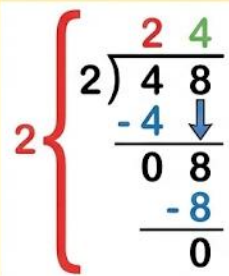
If you know your multiplication facts, you can solve division easily!

If $4 \times 5 = 20$, then:
 $20 \div 5 = 4$ and $20 \div 4 = 5$.

These four numbers (4, 5, 20, $\div$, $\times$) are a fact family.



DIVISION WITHOUT REMAINDER



Steps:

- Tens: $4 \div 2 = 2$. Write 2 above the 4. Multiply: $2 \times 2 = 4$. Subtract: $4 - 4 = 0$.
- Bring down the Ones digit (8).
- Ones: $8 \div 2 = 4$. Write 4 above the 8. Multiply: $4 \times 2 = 8$. Subtract: $8 - 8 = 0$.
- Answer: 24.

DIVISION WITH REMAINDER



Sometimes, numbers don't share perfectly. What is left over is called the Remainder. Example: $13 \div 3$.

Think: How many groups of 3 are in 13?
Count: 3, 6, 9, 12...
The next is 15 (too big).

So, 3 goes into 13 4 times (which makes 12).
 $13 - 12 = 1$ left over.
We write: $13 \div 3 = 4$ remainder 1, or $4 \text{ R } 1$.

Understanding Division

Division means splitting a total into equal parts. The symbol is $\div$. It answers two main questions:

1. **Sharing:** How many does each person get? (e.g., Share 10 sweets between 2 friends: $10 \div 2 = 5$ each).
2. **Grouping:** How many groups can we make? (e.g., How many groups of 3 are in 12? $12 \div 3 = 4$ groups).

Division as Inverse of Multiplication

Division is the exact opposite of multiplication. If you know your multiplication facts, you can solve division easily!

- If $4 \times 5 = 20$, then:
 - $20 \div 5 = 4$
 - $20 \div 4 = 5$
- These four numbers (4, 5, 20, $\div$, $\times$) are a **fact family**.

Division Without Remainder

We can divide digit by digit, starting from the left (the biggest place value).

- **Example:** $48 \div 2$

$$\begin{array}{r} 24 \\ 2 \overline{) 48} \\ \underline{-4} \\ 08 \\ \underline{-8} \\ 0 \end{array}$$

Steps:

- Tens:** $4 \div 2 = 2$. Write 2 above the 4. Multiply: $2 \times 2 = 4$. Subtract: $4 - 4 = 0$.
- Bring down** the Ones digit (8).
- Ones:** $8 \div 2 = 4$. Write 4 above the 8. Multiply: $4 \times 2 = 8$. Subtract: $8 - 8 = 0$.
- Answer: 24.**

Division With Remainder

Sometimes, numbers don't share perfectly. What is left over is called the **Remainder**.

- **Example:** $13 \div 3$

- Think: How many groups of 3 are in 13? Count: 3, 6, 9, 12... The next is 15 (too big).
- So, 3 goes into 13 **4 times** (which makes 12).
- $13 - 12 = 1$ left over.
- We write: **$13 \div 3 = 4$ remainder 1, or 4 R 1.**

EVALUATION

1. What is 20 shared equally among 4 people?
2. If $6 \times 3 = 18$, what is $18 \div 6$?
3. Solve: $84 \div 4$.
4. Solve: $17 \div 5$. (Write the answer with the remainder).
5. Divide 25 by 5.

ASSIGNMENT

1. **Written:** Complete the fact family: $5 \times 4 = 20$, so $20 \div 5 = \underline{\quad}$ and $20 \div 4 = \underline{\quad}$.
2. **Written:** Calculate: $69 \div 3$.
3. **Practical/Oral:** Get 15 small objects (e.g., seeds, pebbles). Try to share them equally among 4 cups. How many are in each cup? How many are left over? This is your remainder.

WEEK 3: DIVISION (PART 2)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Perform long division of 3-digit numbers by 1-digit numbers.
2. Check division answers using multiplication.
3. Solve word problems involving division in real-life contexts.
4. Solve mixed operation problems (multiplication and division).

CONTENT

LONG DIVISION METHOD

Mnemonic: **Divide, Multiply, Subtract, Bring down**

Divide (Daddy) **Multiply (Mummy)** **Subtract (Sister)** **Bring down (Brother)**

Example: $462 \div 2$

$$\begin{array}{r} 231 \\ 2 \overline{) 462} \\ \underline{-4} \\ 06 \\ \underline{-6} \\ 02 \\ \underline{-2} \\ 0 \end{array}$$

a. Divide: $4 \div 2 = 2$ (Write 2)
b. Multiply: $2 \times 2 = 4$
c. Subtract: $4 - 4 = 0$
d. Bring down 6
e. Repeat: $6 \div 2 = 3$, $3 \times 2 = 6$
 $6 - 6 = 0$
Bring down 2
f. Final: $2 \div 2 = 1$, $1 \times 2 = 2$, $2 - 2 = 0$
g. Answer: 231

WORD PROBLEMS

Keywords: 'share equally', 'each', 'divide among', 'per', 'split into groups'

Example: 120 oranges packed equally into 4 baskets.
Think: Total $\div$ Baskets.
Calculation: $120 \div 4 = 30$.
Answer: 30 oranges per basket.

CHECKING DIVISION

Multiply quotient by divisor.
Add remainder if any.
Check for above:
 $231 \times 2 = 462$.
✓ Correct.

Check with remainder:
If $17 \div 5 = 3 \text{ R } 2$, then
 $(3 \times 5) + 2 = 15 + 2 = 17$.
✓ Correct.

MIXED OPERATIONS

A farmer has 5 baskets.
He puts 6 yams in each.
Shares all among 3 sellers.
How many yams does each get?

a. Multiply to find total yams:
 $5 \times 6 = 30$ yams.
b. Divide to share:
 $30 \div 3 = 10$ yams per seller.

Long Division Method

We use four steps: **Divide, Multiply, Subtract, Bring down**. Remember: **Daddy, Mummy, Sister, Brother**.

- **Example:** $462 \div 2$

$$\begin{array}{r} 231 \\ 2 \overline{) 462} \end{array}$$

$$\begin{array}{r}
 -4 \\
 --- \\
 0\ 6 \\
 -6 \\
 --- \\
 0\ 2 \\
 -2 \\
 --- \\
 0
 \end{array}$$

Steps (DMSB):

- Divide:** $4 \div 2 = 2$. Write 2 on top.
- Multiply:** $2 \times 2 = 4$. Write it under the 4.
- Subtract:** $4 - 4 = 0$.
- Bring down the next digit (6).**
- Repeat:** $6 \div 2 = 3$. Multiply: $3 \times 2 = 6$. Subtract: $6 - 6 = 0$. Bring down the last digit (2).
- Final:** $2 \div 2 = 1$. Multiply: $1 \times 2 = 2$. Subtract: $2 - 2 = 0$.
- Answer: 231.**

Checking Division

To check if your division is correct, do the inverse: **Multiply the answer (quotient) by the divisor. Then add the remainder if there is one.**

- *Check for above:* $231 \times 2 = 462$. ✓ Correct.
- *Check with remainder:* If $17 \div 5 = 3 \text{ R } 2$, then check: $(3 \times 5) + 2 = 15 + 2 = 17$. ✓ Correct.

Word Problems

Keywords: "share equally," "each," "divide among," "per," "split into groups."

- *Example:* 120 oranges are to be packed equally into 4 baskets. How many oranges go into each basket?
 - Think: Total oranges $\div$ Number of baskets.
 - Calculation: $120 \div 4 = \mathbf{30}$.
 - Answer: **30** oranges per basket.

Mixed Operations

Sometimes, a problem has more than one step. Read carefully to know what to do first.

- *Problem:* A farmer has 5 baskets. He puts 6 yams in each basket. If he then shares all the yams equally among 3 sellers, how many yams does each seller get?
 - a. **First, multiply** to find total yams: $5 \times 6 = \mathbf{30}$ yams.
 - b. **Then, divide** to share: $30 \div 3 = \mathbf{10}$ yams per seller.

EVALUATION

1. Solve $840 \div 2$ using long division.
2. Check your answer for question 1 using multiplication.
3. Divide 369 by 3.
4. A school has 500 pupils. If they are divided into 5 equal houses, how many pupils are in each house?
5. What is $(10 \times 2) \div 5$?

ASSIGNMENT

1. **Written:** Calculate: $642 \div 2$ using long division.
2. **Written:** Check this answer: Is $15 \div 4 = 3$ remainder 3? Show your check using multiplication and addition.
3. **Practical/Oral:** With a parent, share a packet of biscuits (e.g., 12 biscuits) equally among your family members. How many each? Is there a remainder? Write the division sentence.

WEEK 4: ESTIMATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define estimation as finding an approximate answer.
2. Round numbers to the nearest 10, 100, and 1000.
3. Estimate sums (addition) and differences (subtraction).
4. Explain when estimation is useful in real life.

CONTENT

WHAT IS ESTIMATION? ?	ROUNDING NUMBERS (Review & Extend)									
Estimation is making a smart, quick guess that is close to the exact answer. We use it to check if our calculations make sense, or when we don't need the exact number. We use the symbol $\approx$ which means "is approximately equal to." 	<table border="1"><thead><tr><th>To the nearest 10: Look at the Ones digit.</th><th>To the nearest 100: Look at the Tens digit.</th><th>To the nearest 1000: Look at the Hundreds digit.</th></tr></thead><tbody><tr><td>0-4 → Round DOWN. (e.g., 34 $\approx$ 30, 92 $\approx$ 90).</td><td>0-4 → Round DOWN. (e.g., 120 $\approx$ 100, 349 $\approx$ 300).</td><td>0-4 → Round DOWN. (e.g., 3,400 $\approx$ 3,000, 6,100 $\approx$ 6,000).</td></tr><tr><td>5-9 → Round UP. (e.g., 38 $\approx$ 40, 95 $\approx$ 100).</td><td>5-9 → Round UP. (e.g., 150 $\approx$ 200, 878 $\approx$ 900).</td><td>5-9 → Round UP. (e.g., 2,800 $\approx$ 3,000, 7,600 $\approx$ 8,000).</td></tr></tbody></table>	To the nearest 10: Look at the Ones digit.	To the nearest 100: Look at the Tens digit.	To the nearest 1000: Look at the Hundreds digit.	0-4 → Round DOWN. (e.g., 34 $\approx$ 30, 92 $\approx$ 90).	0-4 → Round DOWN. (e.g., 120 $\approx$ 100, 349 $\approx$ 300).	0-4 → Round DOWN. (e.g., 3,400 $\approx$ 3,000, 6,100 $\approx$ 6,000).	5-9 → Round UP. (e.g., 38 $\approx$ 40, 95 $\approx$ 100).	5-9 → Round UP. (e.g., 150 $\approx$ 200, 878 $\approx$ 900).	5-9 → Round UP. (e.g., 2,800 $\approx$ 3,000, 7,600 $\approx$ 8,000).
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ESTIMATING SUMS AND DIFFERENCES	REAL-LIFE ESTIMATION									
<table border="1"><thead><tr><th>Estimating a Sum: Estimate 38 + 53.</th><th>Estimating a Difference: Estimate 695 - 204.</th></tr></thead><tbody><tr><td>Step 1: Round each number. 38 $\approx$ 40, 53 $\approx$ 50.</td><td>Step 1: Round. 695 $\approx$ 700, 204 $\approx$ 200.</td></tr><tr><td>Step 2: Add the rounded numbers: 40 + 50 = 90.</td><td>Step 2: Subtract: 700 - 200 = 500.</td></tr><tr><td>The exact answer is 91. Our estimate (90) is very close!</td><td>Exact answer is 491. Our estimate is close.</td></tr></tbody></table>	Estimating a Sum: Estimate 38 + 53.	Estimating a Difference: Estimate 695 - 204.	Step 1: Round each number. 38 $\approx$ 40, 53 $\approx$ 50.	Step 1: Round. 695 $\approx$ 700, 204 $\approx$ 200.	Step 2: Add the rounded numbers: 40 + 50 = 90.	Step 2: Subtract: 700 - 200 = 500.	The exact answer is 91. Our estimate (90) is very close!	Exact answer is 491. Our estimate is close.	We use estimation every day!  Shopping: Quickly adding prices to see if you have enough money.  Time: Guessing how long a journey will take.  Food: Estimating how much rice to cook for guests.  It helps us spot silly mistakes. If you calculate 25 + 38 and get 500, estimation (30+40=70) tells you that's wrong!	
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The exact answer is 91. Our estimate (90) is very close!	Exact answer is 491. Our estimate is close.									

What is Estimation?

Estimation is making a smart, quick guess that is close to the exact answer. We use it to check if our calculations make sense, or when we don't need the exact number. We use the symbol $\approx$ which means "is approximately equal to."

Rounding Numbers (Review & Extend)

We round numbers to make them simpler to work with.

- **To the nearest 10:** Look at the **Ones** digit.
 - 0-4 → Round **DOWN**. (e.g., $34 \approx 30$, $92 \approx 90$).
 - 5-9 → Round **UP**. (e.g., $38 \approx 40$, $95 \approx 100$).
- **To the nearest 100:** Look at the **Tens** digit.
 - 0-4 → Round **DOWN**. (e.g., $120 \approx 100$, $349 \approx 300$).
 - 5-9 → Round **UP**. (e.g., $150 \approx 200$, $878 \approx 900$).
- **To the nearest 1000:** Look at the **Hundreds** digit.
 - 0-4 → Round **DOWN**. (e.g., $3,400 \approx 3,000$, $6,100 \approx 6,000$).
 - 5-9 → Round **UP**. (e.g., $2,800 \approx 3,000$, $7,600 \approx 8,000$).

Estimating Sums and Differences

Instead of adding or subtracting tricky numbers, round them first to friendly numbers.

- **Estimating a Sum:** Estimate $38 + 53$.
 - Step 1: Round each number. $38 \approx 40$, $53 \approx 50$.
 - Step 2: Add the rounded numbers: $40 + 50 = \mathbf{90}$.
 - The exact answer is 91. Our estimate (90) is very close!
- **Estimating a Difference:** Estimate $695 - 204$.
 - Step 1: Round. $695 \approx 700$, $204 \approx 200$.
 - Step 2: Subtract: $700 - 200 = \mathbf{500}$.
 - Exact answer is 491. Our estimate is close.

Real-Life Estimation

We use estimation every day!

- **Shopping:** Quickly adding prices to see if you have enough money.
- **Time:** Guessing how long a journey will take.
- **Food:** Estimating how much rice to cook for guests.

- It helps us spot silly mistakes. If you calculate $25 + 38$ and get 500, estimation ($30+40=70$) tells you that's wrong!

EVALUATION

1. Round 45 to the nearest 10.
2. Round 329 to the nearest 100.
3. Estimate the sum of 18 and 21.
4. Estimate the difference between 92 and 48.
5. Give one reason why estimation is useful.

ASSIGNMENT

1. **Written:** Round these to the nearest 1000: (a) 4,200 (b) 8,900.
2. **Written:** Estimate the product (multiplication): 11×19 . (*Hint: Round to nearest 10 first*).
3. **Practical/Oral:** Go to the market with an adult (or look at a flyer). Pick two item prices (e.g., ₦1,950 and ₦3,100). Tell the adult the estimated total cost by rounding to the nearest hundred Naira.

WEEK 5: MID-TERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Attempt questions covering topics from Week 1 to Week 4.

CONTENT

- **Assessment:** The teacher will administer a written test covering Multiplication (tables, 2-digit by 1-digit, grid method), Division (with and without remainder, long division), and Estimation (rounding, estimating sums/differences).

EVALUATION

- Grading and corrections.

ASSIGNMENT

- Review all test corrections with your teacher or parent.

WEEK 6: MID-TERM BREAK



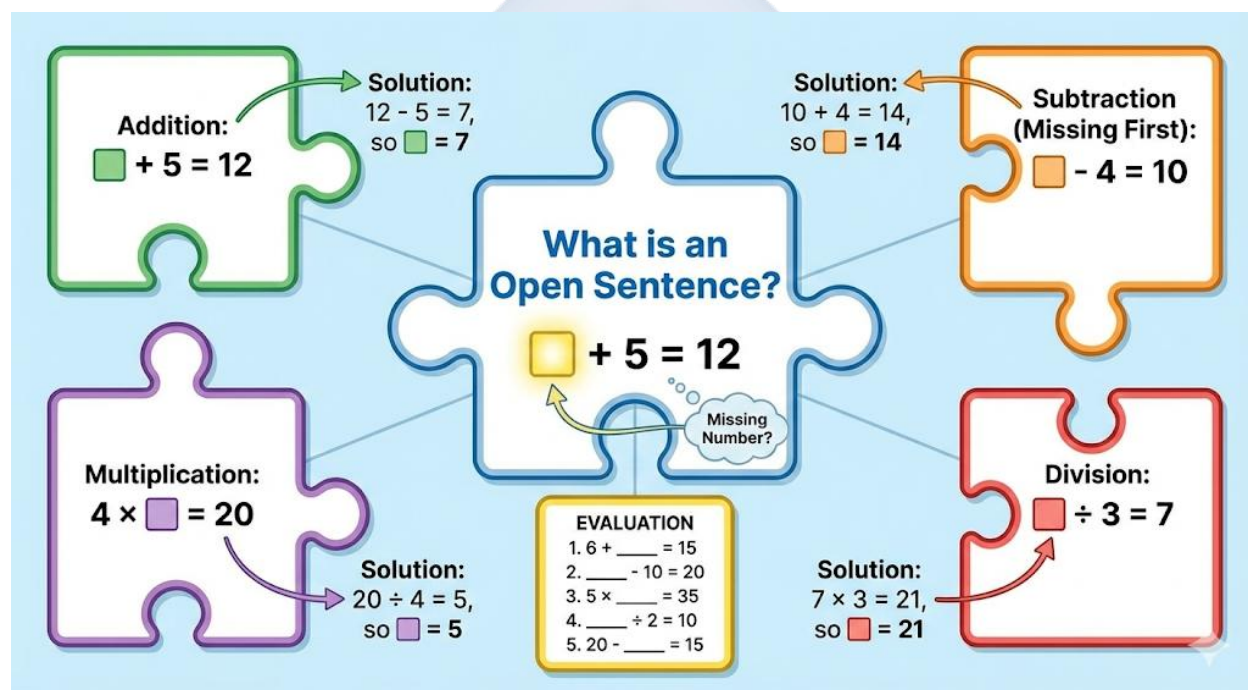
WEEK 7: OPEN SENTENCES

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify an open sentence (equation with a missing number).
2. Find missing numbers in addition and subtraction problems.
3. Find missing numbers in multiplication and division problems.
4. Use inverse operations to solve open sentences.

CONTENT



What is an Open Sentence?

An open sentence is a mathematical statement that is not complete; it has a missing number. The missing number is often shown as a box $\square$, a blank line $\underline{\hspace{1cm}}$, or a letter like **n**. Our job is to find the number that makes the sentence true.

Solving Addition Open Sentences

To find a missing number in an addition sentence, use subtraction (the inverse of addition).

- **Example:** $\square + 5 = 12$.
 - Think: "What number plus 5 gives 12?"
 - Solution: $12 - 5 = 7$. So, $\square = 7$.

Solving Subtraction Open Sentences

Be careful! There are two types. Use the inverse (addition) to solve.

1. **Missing First Number:** $\square - 4 = 10$.
 - Think: "What number, when I take away 4, leaves 10?" The missing number must be **bigger**.
 - Solution: $10 + 4 = 14$. So, $\square = 14$.
2. **Missing Second Number:** $15 - \square = 8$.
 - Think: "15 minus what number gives 8?"
 - Solution: $15 - 8 = 7$. So, $\square = 7$.

Solving Multiplication and Division Open Sentences

- **Multiplication:** $4 \times \square = 20$.
 - Think: "4 groups of what number make 20?" Use division.
 - Solution: $20 \div 4 = 5$. So, $\square = 5$.
- **Division:** $\square \div 3 = 7$.
 - Think: "What number, when shared into 3 groups, gives 7 in each group?" Use multiplication.
 - Solution: $7 \times 3 = 21$. So, $\square = 21$.

EVALUATION

1. Find the missing number: $6 + \underline{\quad} = 15$.
2. Find the missing number: $\underline{\quad} - 10 = 20$.
3. Solve: $5 \times \underline{\quad} = 35$.
4. Solve: $\underline{\quad} \div 2 = 10$.

5. If $20 - \underline{\quad} = 15$, what is the missing number?

ASSIGNMENT

1. **Written:** Complete the open sentence: $50 + \square = 100$.
2. **Written:** Complete: $\square \times 8 = 24$.
3. **Practical/Oral:** Create your own open sentence for a family member (e.g., "I'm thinking of a number. If I add 10 to it, I get 25."). Ask them to find your number.



WEEK 8: MONEY (PART 1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify Nigerian coins and bank notes.
2. State the relationship between Naira and Kobo ($\text{₦}1 = 100\text{k}$).
3. Convert money from Naira to Kobo and Kobo to Naira.
4. Add and subtract amounts of money.

CONTENT

UNDERSTANDING NIGERIAN CURRENCY

We use **Naira (₦)** and **Kobo (k)**. Knowing the different notes and coins helps us buy things and get the right change.

COMMON NOTES & COINS



50k ₦1 ₦2

The Relationship: $\text{₦}1 = 100\text{k}$

CONVERSION

NAIRA TO KOBO

Multiply by 100
(Move decimal two places right, or add zeros)

Examples:
 $\text{₦}5 = 5 \times 100 = 500\text{k}$
 $\text{₦}2.50 = 2.50 \times 100 = 250\text{k}$

KOBO TO NAIRA

Divide by 100
(Move decimal two places left)

Examples:
 $300\text{k} = 300 \div 100 = \text{₦}3.00$
 $150\text{k} = 150 \div 100 = \text{₦}1.50$

MAKING CHANGE

Real-life skill. It is subtraction.

Example:
Buy sweets for $\text{₦}80$.
You give the seller $\text{₦}100$.
Change = Money given – Price of item.
Calculation: $\text{₦}100 - \text{₦}80 = \text{₦}20$.
You should get $\text{₦}20$ change.

ADDITION & SUBTRACTION OF MONEY

Always write money amounts with Naira and Kobo in columns, lining up decimal points.

Example: Add $\text{₦}25.50$ and $\text{₦}10.20$.

₦	k
25	50
+ 10	20
35	70

Add Kobo: $50\text{k} + 20\text{k} = 70\text{k}$.
Add Naira: $\text{₦}25 + \text{₦}10 = \text{₦}35$.
Answer: $\text{₦}35.70$.

Nigerian Currency

We use Naira (₦) and Kobo (k) as our money. Knowing the different notes and coins helps us buy things and get the right change.

- **Common Notes:** ₦5, ₦10, ₦20, ₦50, ₦100, ₦200, ₦500, ₦1000.
- **Common Coins:** 50k, ₦1, ₦2.

The Relationship: ₦1 = 100k

This is the most important rule. One Naira is the same as one hundred Kobo.

Conversion

- **Naira to Kobo: Multiply by 100** (Move the decimal point two places to the right, or add two zeros).
 - Example: ₦5 = $5 \times 100 = 500\text{k}$.
 - Example: ₦2.50 = $2.50 \times 100 = 250\text{k}$.
- **Kobo to Naira: Divide by 100** (Move the decimal point two places to the left).
 - Example: $300\text{k} = 300 \div 100 = \text{₦}3.00$.
 - Example: $150\text{k} = 150 \div 100 = \text{₦}1.50$.

Addition and Subtraction of Money

Always write money amounts with the Naira and Kobo in columns, lining up the decimal points.

- **Example:** Add ₦25.50 and ₦10.20.

₦	k
25	.50
+ 10	.20

35	.70

Steps: Add Kobo: $50\text{k} + 20\text{k} = 70\text{k}$. Add Naira: $\text{₦}25 + \text{₦}10 = \text{₦}35$. Answer: **₦35.70**.

Making Change

This is a very important real-life skill. It is simply subtraction.

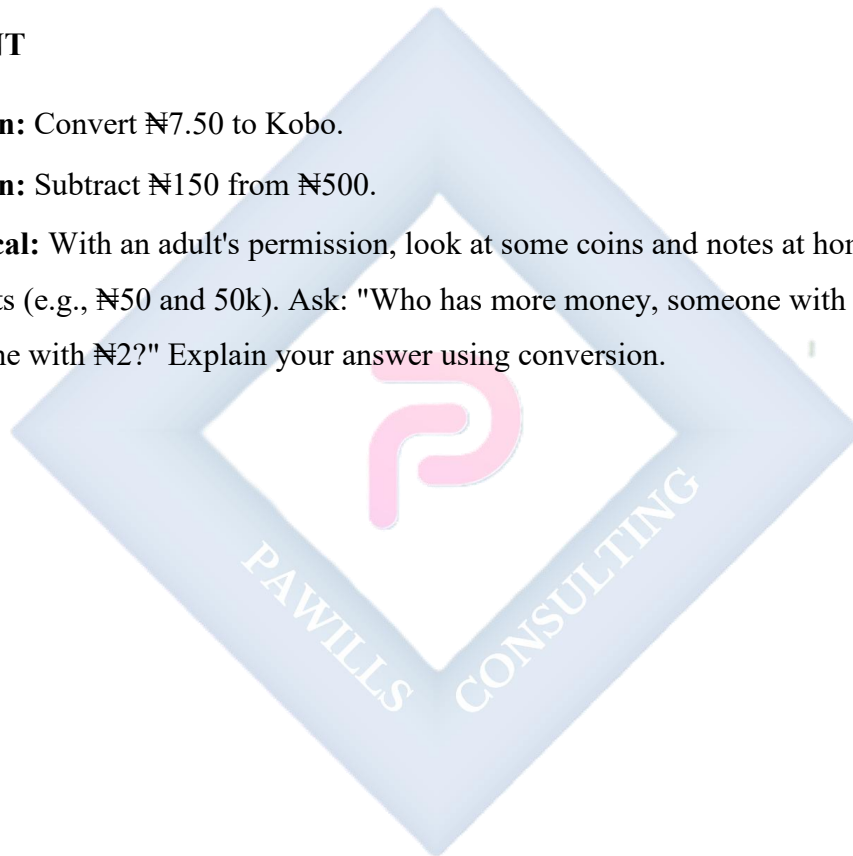
- **Example:** You buy sweets for ₦80. You give the seller ₦100.
 - Change = Money given - Price of item.
 - Calculation: $\text{₦}100 - \text{₦}80 = \text{₦}20$.
 - You should get **₦20** change.

EVALUATION

1. How many Kobo make 1 Naira?
2. Convert ₦4.00 to Kobo.
3. Convert 250k to Naira.
4. Add: ₦50.00 + ₦20.50.
5. If you buy a biscuit for ₦80 and pay with a ₦100 note, how much change should you receive?

ASSIGNMENT

1. **Written:** Convert ₦7.50 to Kobo.
2. **Written:** Subtract ₦150 from ₦500.
3. **Practical:** With an adult's permission, look at some coins and notes at home. Pick two amounts (e.g., ₦50 and 50k). Ask: "Who has more money, someone with 300k or someone with ₦2?" Explain your answer using conversion.



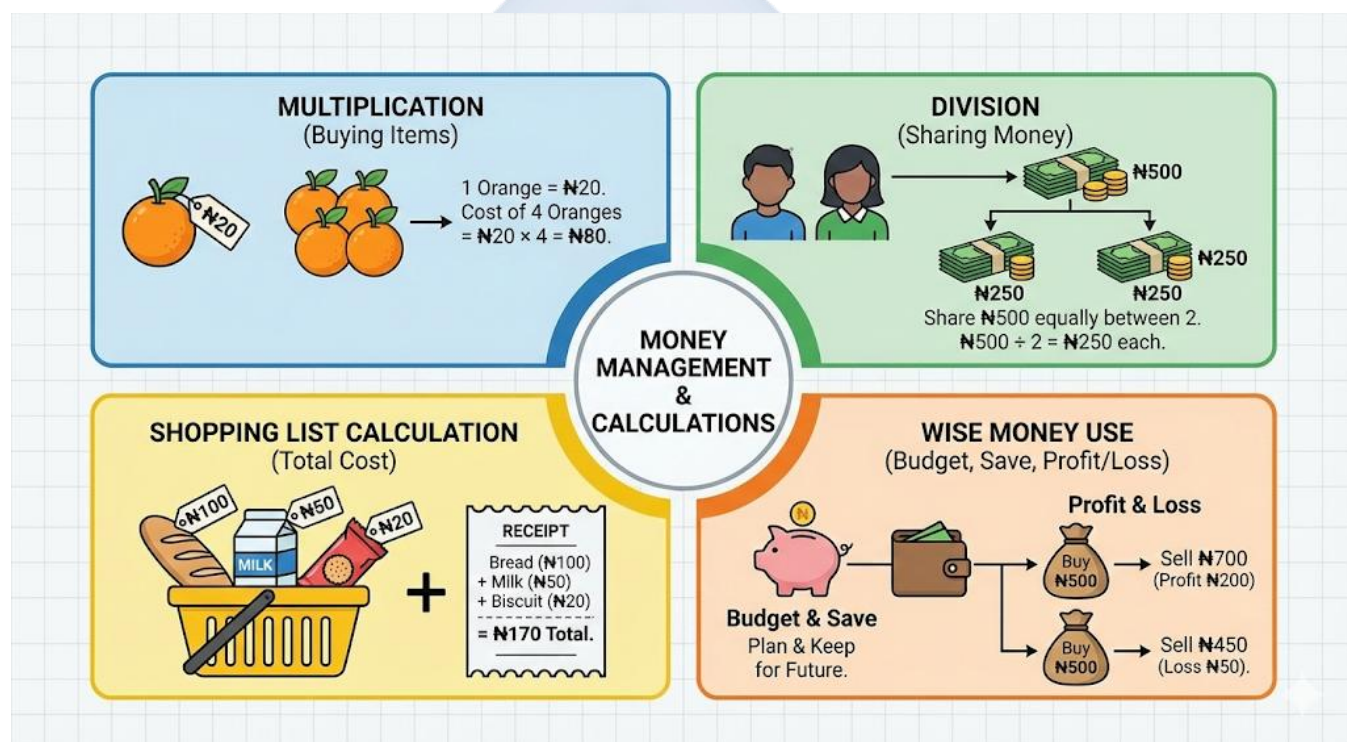
WEEK 9: MONEY (PART 2)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Multiply amounts of money by a whole number.
2. Divide amounts of money by a whole number.
3. Calculate the total cost of a shopping list.
4. Explain simple concepts of profit, loss, and saving.

CONTENT



Multiplication of Money

We multiply money when we buy several of the same item.

- *Example:* If one orange costs ₦20, what is the cost of 4 oranges?
 - Think: **Cost of one $\times$ Number of items.**
 - Calculation: $₦20 \times 4 = ₦80$.

Division of Money

We divide money when we share an amount equally.

- *Example:* Share ₦500 equally between 2 people.
 - Think: **Total money ÷ Number of people.**
 - Calculation: $₦500 \div 2 = ₦250$ each.

Shopping List Calculation

To find the total cost of several different items, we add their prices together.

- *Example:* Shopping List: Bread ₦100, Milk ₦50, Biscuit ₦20.
 - Total Cost = $₦100 + ₦50 + ₦20 = ₦170$.

Budgeting, Saving, Profit and Loss

These are important ideas for wise use of money.

- **Budget:** A plan for how to spend your money so you don't run out. (e.g., I have ₦500, I will spend ₦300 on food and save ₦200).
- **Saving:** Keeping some money for future use instead of spending it all now. Banks keep it safe and may give you extra money called **interest**.
- **Profit:** You make a profit when you sell something for **more** than you bought it.
 - *Example:* Buy a bag for ₦500, sell it for ₦700. Profit = $₦700 - ₦500 = ₦200$.
- **Loss:** You make a loss when you sell something for **less** than you bought it.
 - *Example:* Buy a bag for ₦500, sell it for ₦450. Loss = $₦500 - ₦450 = ₦50$.

EVALUATION

1. Multiply: $₦50 \times 3$.
2. Divide: $₦100 \div 4$.
3. A pen costs ₦20. How much do 5 pens cost?
4. If I buy a toy for ₦500 and sell it for ₦700, did I make a profit or a loss? How much?
5. What does it mean to save money?

ASSIGNMENT

1. **Written:** If one egg costs ₦50, calculate the cost of 10 eggs.
2. **Written:** Share ₦1,000 equally among 5 students. How much does each get?
3. **Practical:** Make a simple budget for your pocket money for one week (e.g., ₦200). Plan how much to spend on snacks and how much to save. Explain your plan to a parent.



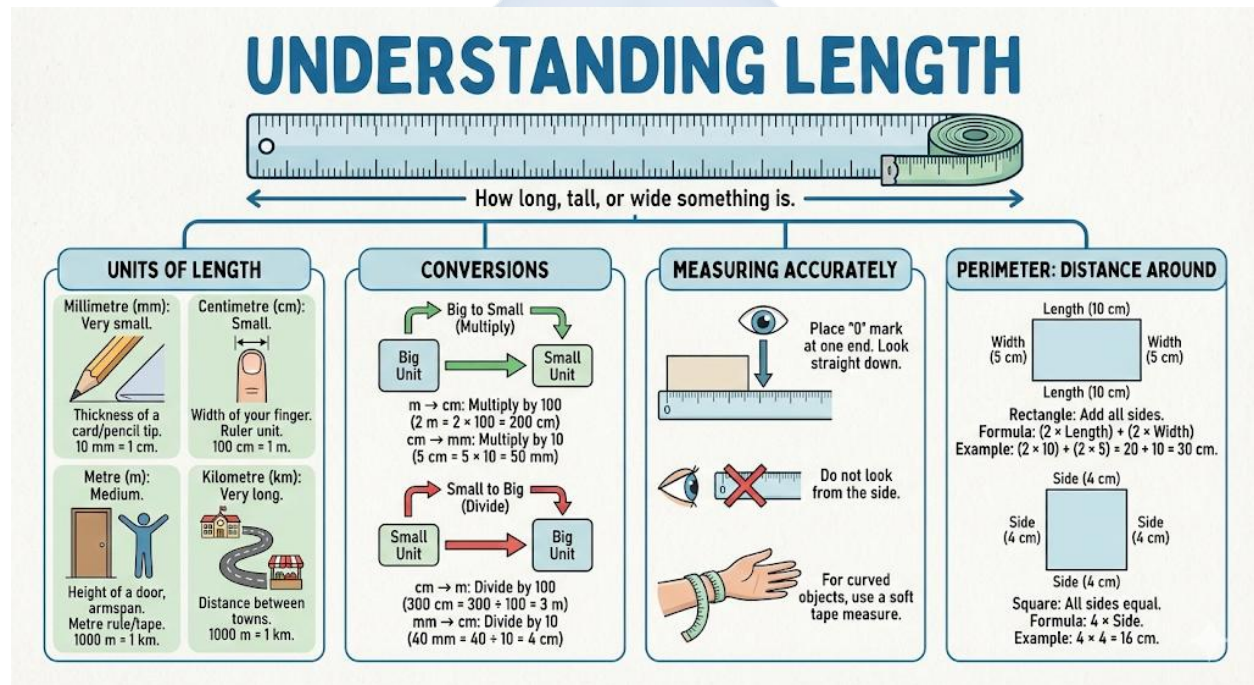
WEEK 10: LENGTH

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify units of length (mm, cm, m, km).
2. Convert between units (e.g., cm to m).
3. Measure objects using a ruler or tape measure.
4. Calculate the perimeter of simple shapes (square, rectangle).

CONTENT



Understanding Length

Length is how long, tall, or wide something is. We measure it using standard units so everyone gets the same answer.

Units of Length

- **Millimetre (mm):** Very, very small. The thickness of a card or the tip of your pencil. **10 mm = 1 cm.**

- **Centimetre (cm):** Small. The width of your finger. We use a ruler marked in cm. **100 cm = 1 m.**
- **Metre (m):** Medium. The height of a door, or your armspan. We use a metre rule or tape measure.
- **Kilometre (km):** Very long. The distance from your school to the market, or between towns. **1000 m = 1 km.**

Conversions

- **Big to Small (Multiply):** To change a bigger unit to a smaller one, you multiply.
 - m → cm: Multiply by 100. ($2\text{ m} = 2 \times 100 = \mathbf{200\text{ cm}}$).
 - cm → mm: Multiply by 10. ($5\text{ cm} = 5 \times 10 = \mathbf{50\text{ mm}}$).
- **Small to Big (Divide):** To change a smaller unit to a bigger one, you divide.
 - cm → m: Divide by 100. ($300\text{ cm} = 300 \div 100 = \mathbf{3\text{ m}}$).
 - mm → cm: Divide by 10. ($40\text{ mm} = 40 \div 10 = \mathbf{4\text{ cm}}$).

Measuring Length Accurately

- Place the '**0**' **mark** of your ruler at one end of the object.
- Look straight down at the other end to see the measurement. Do not look from the side.
- For curved objects (like a wrist), use a soft tape measure.

Perimeter

Perimeter is the **total distance around the outside** of a flat shape. You find it by **adding the lengths of all the sides**.

- **Rectangle:** Perimeter = Length + Width + Length + Width. A shorter way: **(2 × Length) + (2 × Width)**.
 - *Example:* A book is 10 cm long and 5 cm wide.
 - Perimeter = $10 + 5 + 10 + 5 = \mathbf{30\text{ cm}}$.
- **Square:** All sides are equal. Perimeter = **Side + Side + Side + Side**, or **4 × Side**.
 - *Example:* A square tile has side 4 cm.

- Perimeter = $4 + 4 + 4 + 4 = 16 \text{ cm}$, or $4 \times 4 = 16 \text{ cm}$.

EVALUATION

1. How many centimetres make 1 metre?
2. Which unit would you use to measure the distance from Lagos to Abuja? (mm, cm, m, km)
3. Convert 30 mm to cm.
4. Use your ruler to measure the length of your pencil in cm.
5. Calculate the perimeter of a square with a side of 4 cm.

ASSIGNMENT

1. **Written:** Convert 500 cm to metres.
2. **Practical:** At home, use a ruler or tape measure to find the length and width of your mattress or a big table. Write them down.
3. **Drawing/Written:** Draw a rectangle in your book. Label the length 8 cm and the width 2 cm. Calculate and write its perimeter.

WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts from the entire term.
2. Solve mixed problems on Multiplication, Division, Money, and Length.

CONTENT

- **Review – Multiplication:** Tables (6-12). 2-digit $\times$ 1-digit (with carrying). Grid method for 2-digit $\times$ 2-digit.
- **Review – Division:** Meaning (sharing/grouping). Relationship with multiplication. Long division (DMSB). Remainders.
- **Review – Open Sentences:** Finding missing numbers using inverse operations (+, -, $\times$, $\div$).
- **Review – Money:** ₦1 = 100k. Conversion. Addition, subtraction, multiplication, division of money. Profit/Loss.
- **Review – Length:** Units (mm, cm, m, km). Conversions. Measuring. Perimeter of rectangles and squares.

EVALUATION

- Oral quiz and written revision exercises covering the term's work (e.g., "What is 12×9 ?", "Share ₦450 among 3 people", "Find the perimeter of a 7cm by 3cm rectangle").

ASSIGNMENT

- Study all notes for the end-of-term examination. Practice 2 problems from each of the 5 main topics (Multiplication, Division, Open Sentences, Money, Length).

PRIMARY 4 MATHEMATICS SCHEME OF WORK

THIRD TERM

Week	Topic	Breakdown (Subtopics)
1	Weight	• Understanding weight: how heavy something is • Units of weight: gram (g), kilogram (kg) • Conversion: 1000g = 1kg • Measuring weight using scales (kitchen, bathroom, market scales) • Estimating weights • Comparing weights: heavier, lighter, same • Addition and subtraction of weights • Word problems on weight
2	Time	• Telling time: o'clock, half past, quarter past, quarter to • 12-hour and 24-hour clock (introduction) • a.m. and p.m. • Reading timetables and schedules • Calculating duration: how long activities take • Days, weeks, months, years • Calendar activities: finding dates, counting days • Word problems on time
3	Area	• Understanding area: space inside a shape • Measuring area using square units (square cm, square m) • Counting squares to find area • Area of rectangles: length $\times$ width • Area of squares: side $\times$ side • Comparing areas of different shapes • Practical activities: measuring classroom, desk,

		book areas • Word problems on area
4	Capacity	• Understanding capacity: how much a container holds • Units of capacity: milliliter (ml), liter (l) • Conversion: 1000ml = 1l • Measuring capacity using measuring cups, jugs, bottles • Estimating capacity • Addition and subtraction of capacity • Real-life applications: cooking, filling containers • Word problems on capacity
5	MID-TERM EXAMINATION	• Assessment of Weeks 1-4
6	MID-TERM BREAK	• Holiday/Rest
7	Plane Shapes (2D Shapes)	• Identifying and naming 2D shapes: circle, triangle, square, rectangle, pentagon, hexagon, octagon • Properties of 2D shapes: sides, corners (vertices), angles • Regular and irregular shapes • Lines of symmetry in shapes • Drawing and constructing 2D shapes • Tessellation: shapes that fit together • Finding shapes in the environment
8	Three Dimensional Shapes (3D Shapes)	• Identifying and naming 3D shapes: cube, cuboid, sphere, cylinder, cone, pyramid • Properties of 3D shapes: faces, edges, vertices

		- Relationship between 2D and 3D shapes (faces of 3D shapes are 2D) - Nets of 3D shapes (folding flat patterns to make shapes) - Finding 3D shapes in the environment - Building and modeling 3D shapes
9	Bar Graphs (Part 1)	- Understanding data: information we collect - Collecting data: surveys, tallying, counting - Organizing data in tables - Introduction to bar graphs - Reading bar graphs: interpreting information - Parts of a bar graph: title, labels, scale, bars - Answering questions from bar graphs
10	Bar Graphs (Part 2)	- Constructing simple bar graphs (vertical and horizontal) - Choosing appropriate scales - Creating bar graphs from collected data - Comparing data using bar graphs - Real-life applications: class surveys (favorite food, sport, color) - Pictographs (introduction): using pictures to represent data - Interpreting and creating pictographs
11	REVISION	Review: Weight, time, area, capacity, shapes, bar graphs
12	EXAMINATION	End of Term Assessment
13	CLOSING/VACATION	Term ends

PRIMARY 4 MATHEMATICS LESSON NOTES

THIRD TERM

WEEK 1: WEIGHT

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define weight as how heavy or light an object is.
2. Identify units of weight (gram and kilogram) and state the relationship ($1000\text{g} = 1\text{kg}$).
3. Read measurements from different types of scales.
4. Add and subtract weights in grams and kilograms.

CONTENT

Understanding Weight

Weight is how heavy something is. It is the force pulling an object towards the Earth.

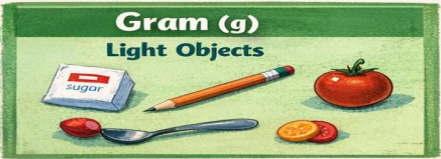
Light: Feather

Heavy: Bag of Books

Units of Weight

Gram (g)

Light Objects



Kilogram (kg)

Heavy Objects



Key Relationship: $1000\text{g} = 1\text{kg}$

$1\text{kg} = 1000\text{g}$ $1000\text{ grams} = 1\text{ Kilogram}$ $3000\text{g} \div 1000 = 3\text{kg}$

Kitchen Scale

Bathroom Scale

Market Scale

Addition & Subtraction of Weights

Example 1: Addition

kg	g
2	500
+	1 200
3 700g	

$2\text{ kg } 500\text{ g} + 1\text{ kg } 200\text{ g} = 3\text{ kg } 700\text{ g}$

Example 2: Subtraction

kg	g
5	400
-	1 100
4 300g	

$5\text{ kg } 400\text{ g} - 1\text{ kg } 100\text{ g} = 4\text{ kg } 300\text{ g}$

Understanding Weight

Weight tells us **how heavy** something is. It is the measure of the force pulling an object down towards the earth. We can feel weight when we lift things. A feather is **light**, so it has a small weight. A bag of books is **heavy**, so it has a large weight.

Units of Weight

We use standard units to measure and talk about weight clearly. The main units are:

- **Gram (g):** Used for **light** objects.
 - *Examples:* A sachet of sugar, a pencil, a key, a small tomato.
- **Kilogram (kg):** Used for **heavy** objects.
 - *Examples:* A bag of rice, a big yam, a watermelon, your own body weight.

The Key Relationship: 1000g = 1kg

This is the most important fact about weight. It means **one thousand grams make one kilogram**. It is like saying 100 kobo make 1 Naira.

- To change **kg to g**, **multiply by 1000**. (e.g., $2 \text{ kg} = 2 \times 1000 = 2000 \text{ g}$).
- To change **g to kg**, **divide by 1000**. (e.g., $3000 \text{ g} = 3000 \div 1000 = 3 \text{ kg}$).

Measuring Weight

We use a **scale** or **balance** to measure weight.

- **Kitchen Scale:** Measures food like flour, beans, or meat in grams (g) or kilograms (kg).
- **Bathroom Scale:** Measures people's body weight in kilograms (kg).
- **Market Scale (Balance):** Traders use this to weigh goods. It often has two pans.

When reading a scale, look carefully at the numbers and the unit (g or kg) shown.

Addition and Subtraction of Weights

We add or subtract weights just like numbers, but we must keep the units (kg and g) in separate columns.

- **Example 1: Addition** – Add 2kg 500g and 1kg 200g.

$$\begin{array}{r}
 \text{kg} \quad \text{g} \\
 2 \quad 500 \\
 + 1 \quad 200 \\
 \hline
 3 \quad 700
 \end{array}$$

Answer: **3 kg 700 g.**

- **Example 2: Subtraction** – Subtract 1kg 100g from 5kg 400g.

$$\begin{array}{r}
 \text{kg} \quad \text{g} \\
 5 \quad 400 \\
 - 1 \quad 100 \\
 \hline
 4 \quad 300
 \end{array}$$

Answer: **4 kg 300 g.**

EVALUATION

1. What do we mean by the 'weight' of an object?
2. How many grams are there in 1 kilogram?
3. Would you measure a car in grams or kilograms?
4. Convert 4000g to kg.
5. Add: 3kg 200g + 1kg 100g.
6. A bag of rice weighs 5kg. If you use 2kg for cooking, how much is left?

ASSIGNMENT

1. **Written:** Convert 5kg to grams.
2. **Practical/Oral:** Find two packaged items at home (e.g., milk, sugar). Read their weights. Tell a sibling which one is heavier and by how much (e.g., "The sugar is 500g heavier than the milk").

3. **Written:** Subtract 250g from 900g.



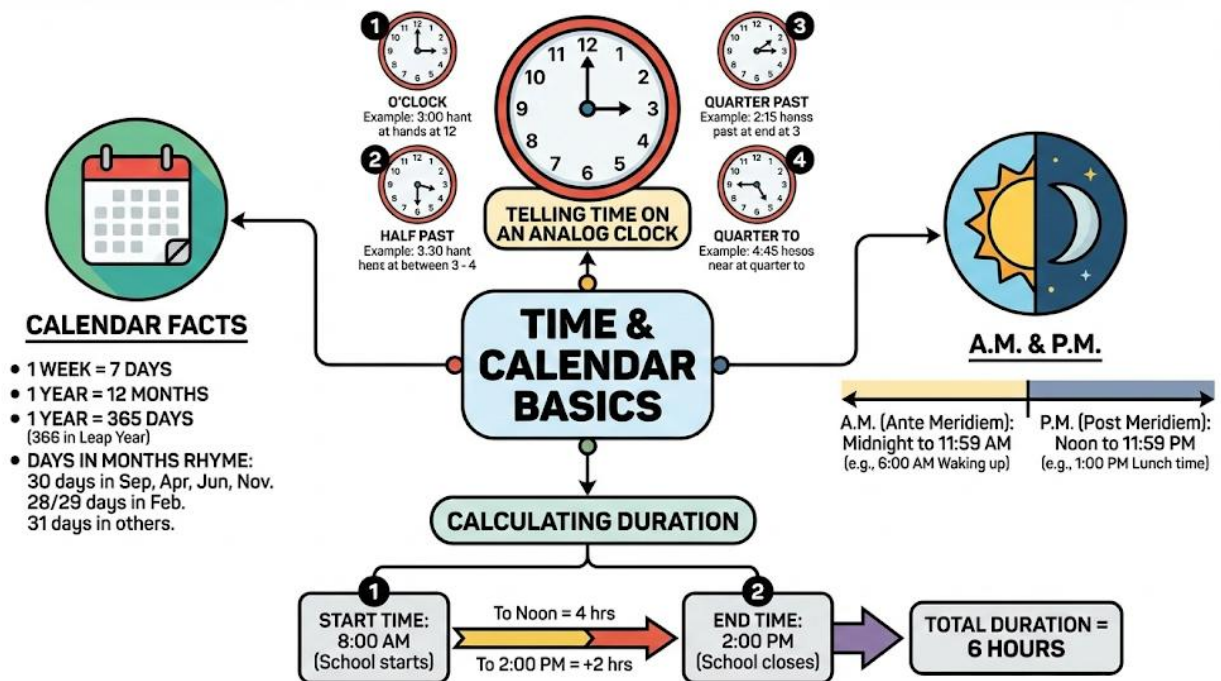
WEEK 2: TIME

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Tell time using "o'clock," "half past," "quarter past," and "quarter to."
2. Differentiate between a.m. (morning) and p.m. (afternoon/evening).
3. Calculate the duration of an activity (how long it takes).
4. Solve simple word problems involving days, weeks, and months.

CONTENT



Telling Time on an Analog Clock

An analog clock has a **short hand (hour hand)** and a **long hand (minute hand)**.

- **O'clock:** When the **long hand points to 12**. The short hand tells the hour.
 - *Example:* Short hand on 3, long hand on 12 = **3 o'clock**.
- **Half Past:** When the **long hand points to 6**. This means 30 minutes have passed.
 - *Example:* Short hand between 3 and 4, long hand on 6 = **Half past 3** or **3:30**.

- **Quarter Past:** When the **long hand points to 3**. This means 15 minutes have passed.
 - *Example:* Short hand just past 2, long hand on 3 = **Quarter past 2** or **2:15**.
- **Quarter To:** When the **long hand points to 9**. This means 15 minutes to the next hour.
 - *Example:* Short hand almost at 5, long hand on 9 = **Quarter to 5** or **4:45**.

A.M. and P.M.

We use these to tell if a time is in the morning or afternoon/evening.

- **a.m. (Ante Meridiem):** From midnight (12:00 at night) until 11:59 in the morning.
 - *Examples:* Waking up (6:00 a.m.), school assembly (8:00 a.m.).
- **p.m. (Post Meridiem):** From noon (12:00 midday) until 11:59 at night.
 - *Examples:* Lunch time (1:00 p.m.), bedtime (9:00 p.m.).

Calculating Duration

Duration is the **length of time** something takes from start to finish. We find it by **subtracting the start time from the end time**.

- *Example:* School starts at 8:00 a.m. and closes at 2:00 p.m.
 - From 8:00 a.m. to 12:00 p.m. (noon) is **4 hours**.
 - From 12:00 p.m. to 2:00 p.m. is **2 hours**.
 - Total duration = 4 hours + 2 hours = **6 hours**.

Calendar Facts

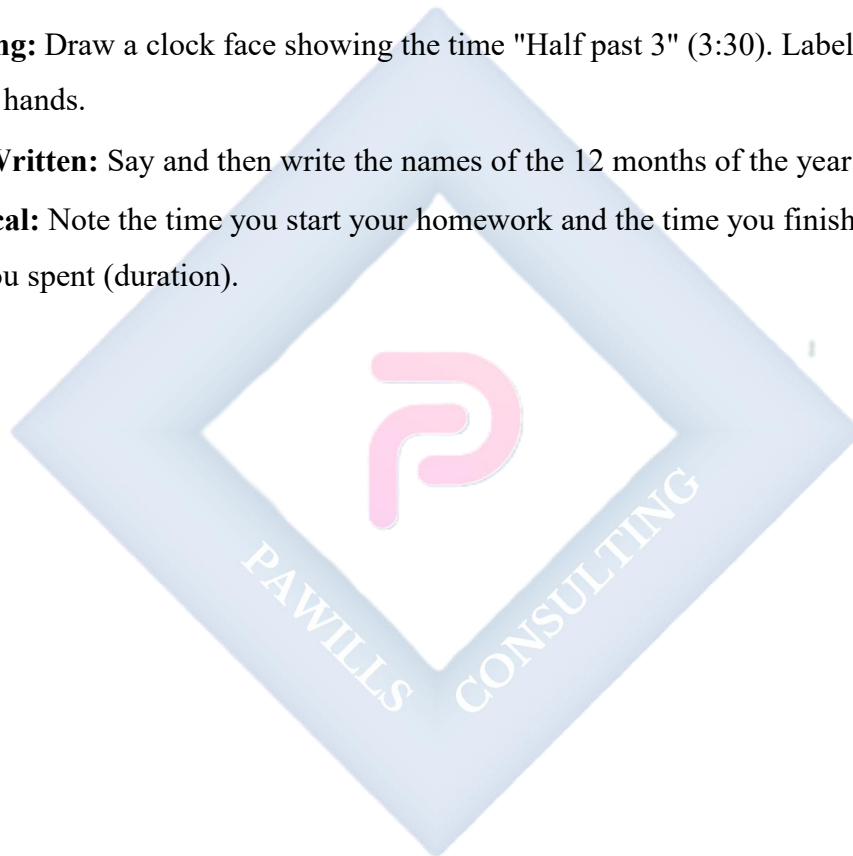
- **1 week = 7 days** (Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday).
- **1 year = 12 months**.
- **1 year = 365 days** (366 days in a **Leap Year**, which comes every 4 years).
- **Days in Months Rhyme:** *Thirty days hath September, April, June, and November. All the rest have thirty-one, Except February alone, Which has twenty-eight days clear, And twenty-nine in each leap year.*

EVALUATION

1. Where is the long (minute) hand when the time is "Half Past" an hour?
2. Do we usually go to school in the a.m. or p.m.?
3. How many days are there in a week? Name them.
4. If a football match starts at 4:00 p.m. and lasts for 1 hour 30 minutes, what time will it end?
5. Which month has the fewest number of days?

ASSIGNMENT

1. **Drawing:** Draw a clock face showing the time "Half past 3" (3:30). Label the hour and minute hands.
2. **Oral/Written:** Say and then write the names of the 12 months of the year in order.
3. **Practical:** Note the time you start your homework and the time you finish. Calculate how long you spent (duration).



WEEK 3: AREA

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define area as the space inside a flat shape.
2. Calculate the area of a rectangle using the formula: $\text{Length} \times \text{Width}$.
3. Calculate the area of a square using the formula: $\text{Side} \times \text{Side}$.
4. Solve simple word problems involving area.

CONTENT

Understanding Area

Area is the amount of flat space a shape covers. Imagine painting the floor of your room. The amount of paint you need depends on the **area** of the floor. Area is measured in **square units**, such as square centimetres (cm^2) or square metres (m^2).



Area of a Rectangle

A rectangle has two opposite sides equal. **Length** (longer side) and **Width** (shorter side).



Formula: **Area** = $\text{Length} \times \text{Width}$.

Example: A whiteboard is 200 cm long and 100 cm wide.
 $\text{Area} = 200 \text{ cm} \times 100 \text{ cm} = 20,000 \text{ cm}^2$.
We say, 'The area is twenty thousand square centimetres.'

Area of a Square

A square is a rectangle where all four sides are equal.

Formula: **Area** = $\text{Side} \times \text{Side}$

Example: A square tile has a side of 30 cm.



Example: A square tile has a side of 30 cm.
 $\text{Area} = 30 \text{ cm} \times 30 \text{ cm} = 900 \text{ cm}^2$.

Why 'Square' Units?

We use '**square**' units because we are counting how many unit squares can fit inside the shape. For example, a rectangle 4 cm by 3 cm can fit 12 squares, each 1 cm by 1 cm. So, its area is 12 square centimetres.



Practical Activity



You can find the **area** of your **notebook** by measuring its **length** and **width** with a ruler (in cm) and then multiplying them.

Understanding Area

Area is the amount of **flat space** a shape covers. Imagine painting the floor of your room. The amount of paint you need depends on the **area** of the floor. Area is measured in **square units**, such as square centimetres (cm²) or square metres (m²).

Area of a Rectangle

A rectangle has two different measurements: **Length (longer side)** and **Width (shorter side)**.

- **Formula: Area = Length × Width**
- *Example:* A whiteboard is 200 cm long and 100 cm wide.
 - Area = 200 cm × 100 cm = **20,000 cm²**.
 - We say, "The area is twenty thousand square centimetres."

Area of a Square

A square is a special rectangle where all four sides are equal.

- Since Length = Width, we call the measurement the **Side**.
- **Formula: Area = Side × Side**
- *Example:* A square tile has a side of 30 cm.
 - Area = 30 cm × 30 cm = **900 cm²**.

Why 'Square' Units? We use 'square' units because we are counting how many unit squares can fit inside the shape. For example, a rectangle 4 cm by 3 cm can fit 12 squares, each 1 cm by 1 cm. So, its area is 12 **square centimetres** (12 cm²).

Practical Activity You can find the area of your notebook by measuring its length and width with a ruler (in cm) and then multiplying them.

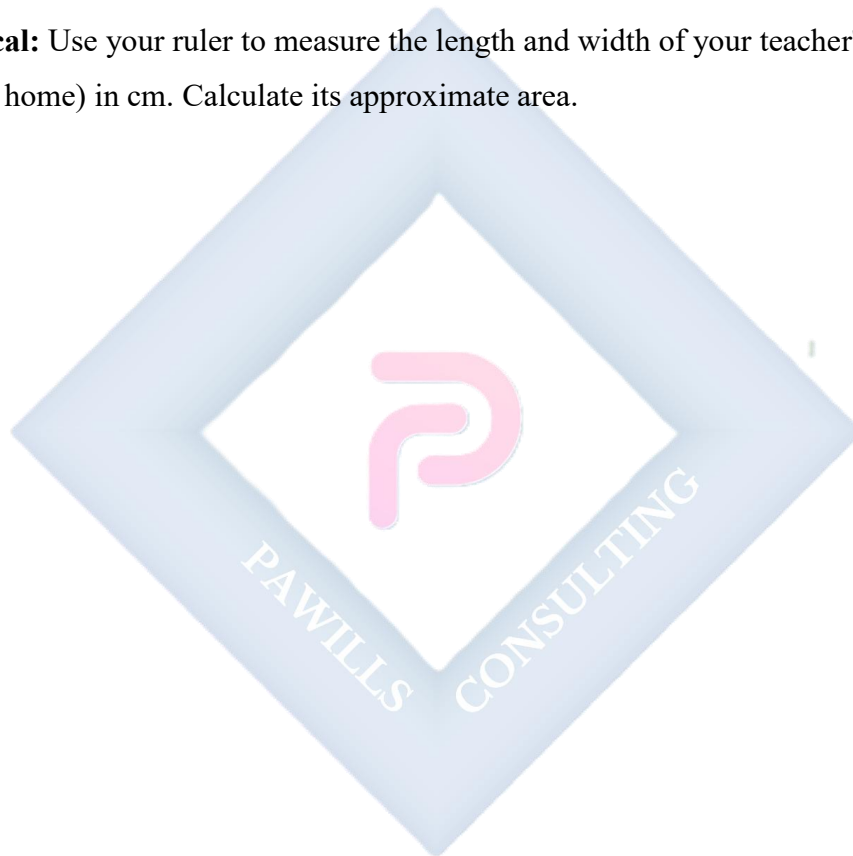
EVALUATION

1. What is area?
2. What is the unit for area? (cm² or m²)
3. Calculate the area of a rectangle with Length 6 cm and Width 2 cm.

4. Find the area of a square with a side of 5 cm.
5. Which classroom floor has a bigger area: one that is 8 m by 5 m, or one that is 7 m by 6 m? (*Hint: Calculate both areas*)

ASSIGNMENT

1. **Written:** Find the area of a school garden that is 10 metres long and 5 metres wide.
2. **Drawing:** Draw a square in your book with a side of 4 cm. Divide it into 1 cm squares. Count the small squares to confirm the area is 16 cm².
3. **Practical:** Use your ruler to measure the length and width of your teacher's table (or your desk at home) in cm. Calculate its approximate area.



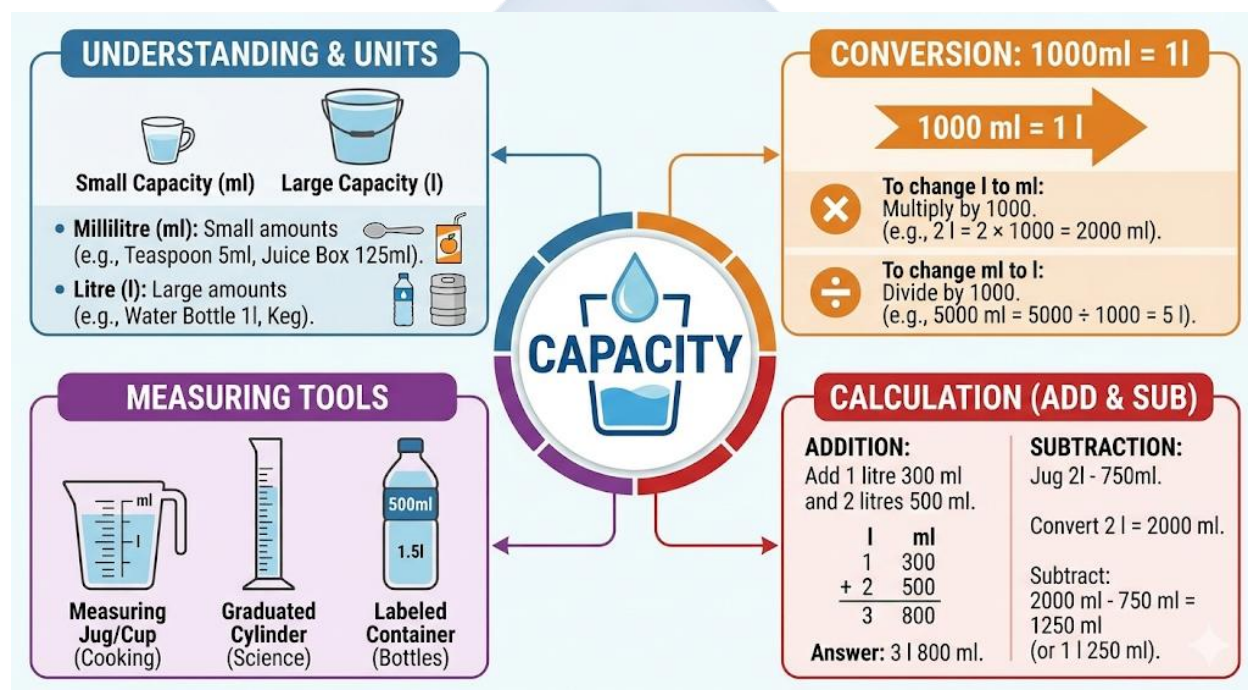
WEEK 4: CAPACITY

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define capacity as the amount of liquid a container can hold.
2. Identify units of capacity: Millilitre (ml) and Litre (l).
3. State the relationship: $1000\text{ml} = 1\text{l}$.
4. Add and subtract capacities in litres and millilitres.

CONTENT



Understanding Capacity

Capacity is the **maximum amount** of liquid (like water, milk, or petrol) that a container can hold. A small cup has a small capacity. A big bucket has a large capacity.

Units of Capacity

- **Millilitre (ml):** Used for **small amounts** of liquid.

- *Examples:* A teaspoon of medicine (5 ml), a small juice box (125 ml), a bottle of eyedrops.
- **Litre (l):** Used for **larger amounts** of liquid.
 - *Examples:* A bottle of soft drink (35 cl is 350ml, but 1 litre is common), a jug of water, a keg of palm wine.

The Key Relationship: 1000ml = 1l

Just like weight, capacity has a key conversion: **One thousand millilitres make one litre.**

- To change **l to ml**, **multiply by 1000**. (e.g., 2 l = $2 \times 1000 = 2000$ ml).
- To change **ml to l**, **divide by 1000**. (e.g., 5000 ml = $5000 \div 1000 = 5$ l).

Measuring Capacity

We use special tools to measure liquids accurately.

- **Measuring Jug/Cup:** Has lines showing millilitres (ml) and litres (l). Used in cooking.
- **Graduated Cylinder:** Used in science labs for precise measurement.
- **Containers with labels:** Milk, water, and juice bottles show their capacity (e.g., 500ml, 1.5l).

Addition and Subtraction of Capacities

We calculate with capacity just like with weight, keeping litres and millilitres in columns.

- **Example 1: Addition** – Add 1 litre 300 ml and 2 litres 500 ml.

$$\begin{array}{r}
 1 \quad \text{ml} \\
 1 \quad 300 \\
 + 2 \quad 500 \\
 \hline
 3 \quad 800
 \end{array}$$

Answer: **3 l 800 ml.**

- **Example 2: Subtraction** – A jug has 2 litres of water. You pour out 750 ml to drink. How much is left?

- First, convert: 2 l = **2000 ml**.
- Subtract: 2000 ml - 750 ml = **1250 ml** (or **1 l 250 ml**).

EVALUATION

1. What does the 'capacity' of a bottle tell us?
2. Which is a larger unit: millilitre or litre?
3. How many millilitres are there in 1 litre?
4. Add: 500 ml + 200 ml.
5. A water dispenser has 5 litres of water. If people drink 2 litres, how many litres are left?

ASSIGNMENT

1. **Written:** Convert 3 litres to millilitres.
2. **Practical/Oral:** Find three different liquid containers at home (e.g., milk, oil, soft drink). Read their capacities aloud and say which holds the most.
3. **Written:** A bottle contains 750 ml of juice. You drink 250 ml. How much juice is left? Give your answer in ml.

WEEK 5: MID-TERM EXAMINATION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Attempt questions covering topics from Week 1 to Week 4.

CONTENT

- **Assessment:** The teacher will administer a written test covering Weight (units, conversion, calculations), Time (telling time, duration, calendar), Area (of rectangles and squares), and Capacity (units, conversion, calculations).

EVALUATION

- Grading and corrections.

ASSIGNMENT

- Review all test corrections with your teacher or parent.

WEEK 6: MID-TERM BREAK



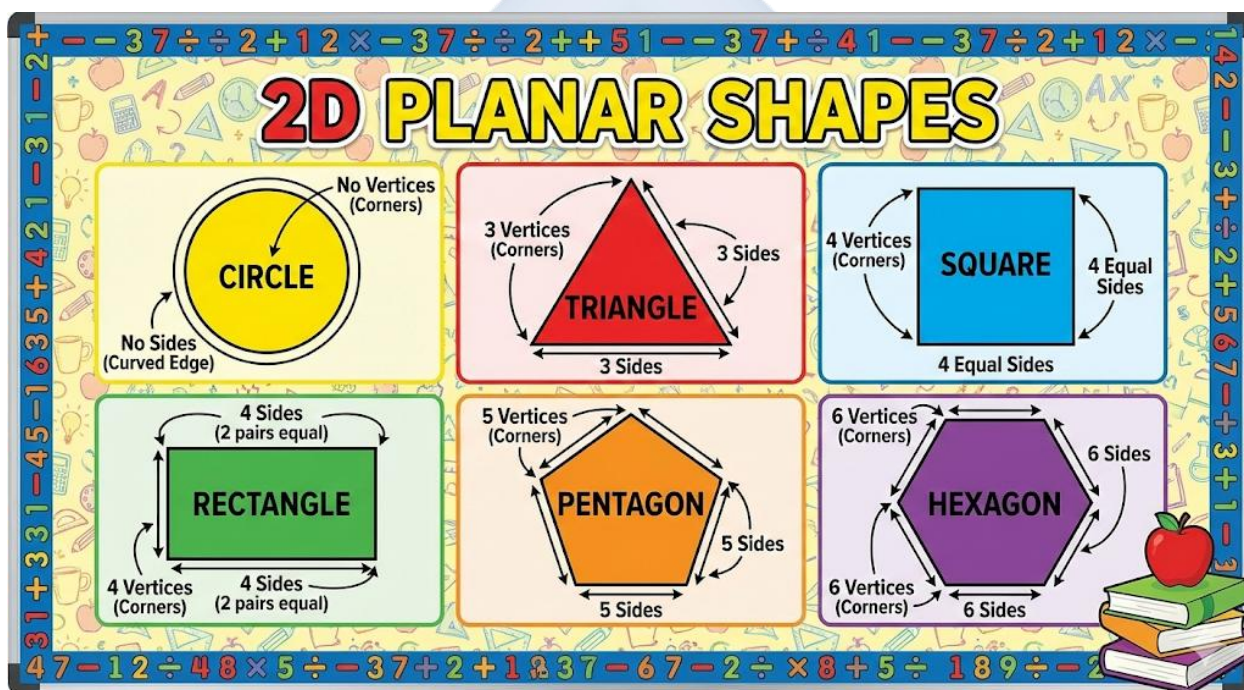
WEEK 7: PLANE SHAPES (2D SHAPES)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify and name 2D shapes (circle, triangle, square, rectangle, pentagon, hexagon).
2. List the properties of 2D shapes (number of sides and corners/vertices).
3. Identify lines of symmetry in shapes.
4. Draw 2D shapes correctly.

CONTENT



Identifying 2D Shapes

2D (Two-Dimensional) shapes are **flat**. They have only **length and width** (like a drawing on paper). They do not have thickness. We see their outlines.

- **Circle:** A round shape with **no straight sides** and **no corners**. Every point on the edge is the same distance from the centre.
- **Triangle:** A shape with **3 straight sides** and **3 corners** (vertices).

- **Square:** A shape with **4 equal sides** and **4 right-angle corners**.
- **Rectangle:** A shape with **4 sides**. Opposite sides are **equal and parallel**. It has **4 right-angle corners**.
- **Pentagon:** A shape with **5 sides** and **5 corners**.
- **Hexagon:** A shape with **6 sides** and **6 corners** (like a honeycomb cell).
- **Octagon:** A shape with **8 sides** and **8 corners** (like a 'STOP' sign).

Properties: Sides and Vertices

- **Side:** A straight line that forms part of the shape's boundary.
- **Vertex (plural: Vertices):** A corner; the point where two sides meet.

We describe a shape by counting these. *Example:* "A triangle has 3 sides and 3 vertices."

Lines of Symmetry

A line of symmetry is like a **mirror line** through a shape. If you fold the shape along this line, both halves match **exactly**.

- A square has **4** lines of symmetry (vertical, horizontal, two diagonals).
- A rectangle has **2** lines of symmetry (vertical and horizontal).
- A circle has **infinite** lines of symmetry (you can draw a line through the centre in any direction).
- A regular pentagon has **5** lines of symmetry.

EVALUATION

1. How many sides does a triangle have?
2. Name a shape that has 5 sides.
3. How many corners (vertices) does a circle have?
4. Draw a square and use a dotted line to show one of its lines of symmetry.
5. What is the difference between a square and a rectangle?

ASSIGNMENT

1. **Drawing:** Draw a hexagon. Label its sides and vertices.
2. **Practical Hunt:** At home or in school, find and list one real-life object for three different 2D shapes (e.g., clock - circle, book - rectangle).
3. **Folding Activity:** Cut out a square and a rectangle from paper. Fold them to find and draw all their lines of symmetry.



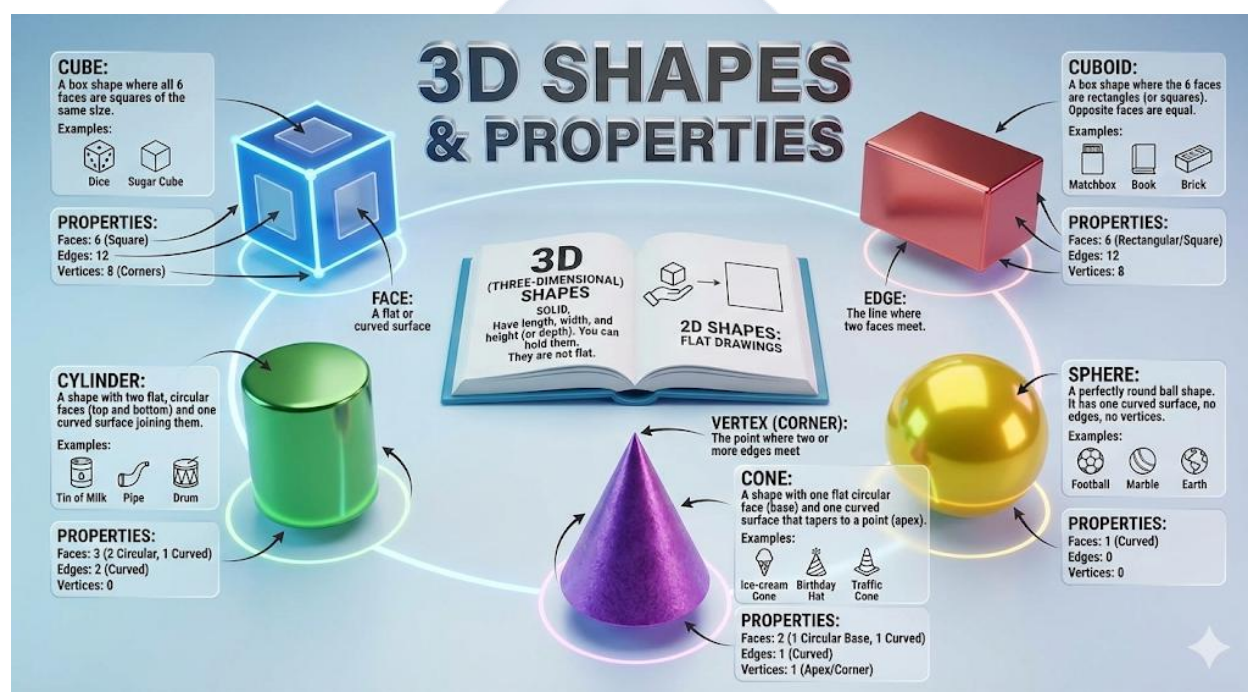
WEEK 8: THREE DIMENSIONAL SHAPES (3D SHAPES)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Identify and name 3D shapes (cube, cuboid, sphere, cylinder, cone).
2. State the properties of 3D shapes (faces, edges, vertices).
3. Differentiate between 2D and 3D shapes.
4. Identify 3D shapes in the environment.

CONTENT



Identifying 3D Shapes

3D (Three-Dimensional) shapes are **solid**. They have **length, width, and height (or depth)**. You can hold them. They are not flat.

- **Cube:** A box shape where all **6 faces are squares** of the same size. *Example:* A dice, a sugar cube.
- **Cuboid:** A box shape where the **6 faces are rectangles** (or squares). Opposite faces are equal. *Example:* A matchbox, a book, a brick.

- **Sphere:** A perfectly round ball shape. It has **one curved surface**, no edges, no vertices.
Example: A football, a marble, the Earth.
- **Cylinder:** A shape with **two flat, circular faces** (top and bottom) and **one curved surface** joining them. *Example:* A tin of milk, a pipe, a drum.
- **Cone:** A shape with **one flat circular face** (base) and **one curved surface** that tapers to a point (apex). *Example:* An ice-cream cone, a birthday hat, a traffic cone.

Properties: Faces, Edges, Vertices

These are the parts that make up a 3D shape.

- **Face:** A flat or curved **surface**.
- **Edge:** The line where **two faces meet**.
- **Vertex (Corner):** The point where **two or more edges meet**.

Let's count for a **Cube**:

- **Faces:** 6 (All are square faces).
- **Edges:** 12 (Each edge is where two squares meet).
- **Vertices:** 8 (Each corner of the cube).

2D vs. 3D: The Main Difference

- **2D shapes** are **flat drawings** (e.g., a drawing of a square on paper).
- **3D shapes** are **solid objects** you can pick up (e.g., a cube-shaped box).

EVALUATION

1. Name a 3D shape that looks like a ball.
2. How many faces does a cube have?
3. What is the difference between a square (2D) and a cube (3D)?
4. Name an object in this classroom that is shaped like a cuboid.
5. How many edges does a sphere have?

ASSIGNMENT

1. **Drawing:** Draw a cylinder. Label its circular faces and curved surface.
2. **Hunt & List:** Find and list two objects at home for each: a sphere and a cylinder.
3. **Model Making:** Using cardboard, old boxes, or moulding clay, try to make one 3D shape (cube, cuboid, or cylinder). Show it to your classmates.



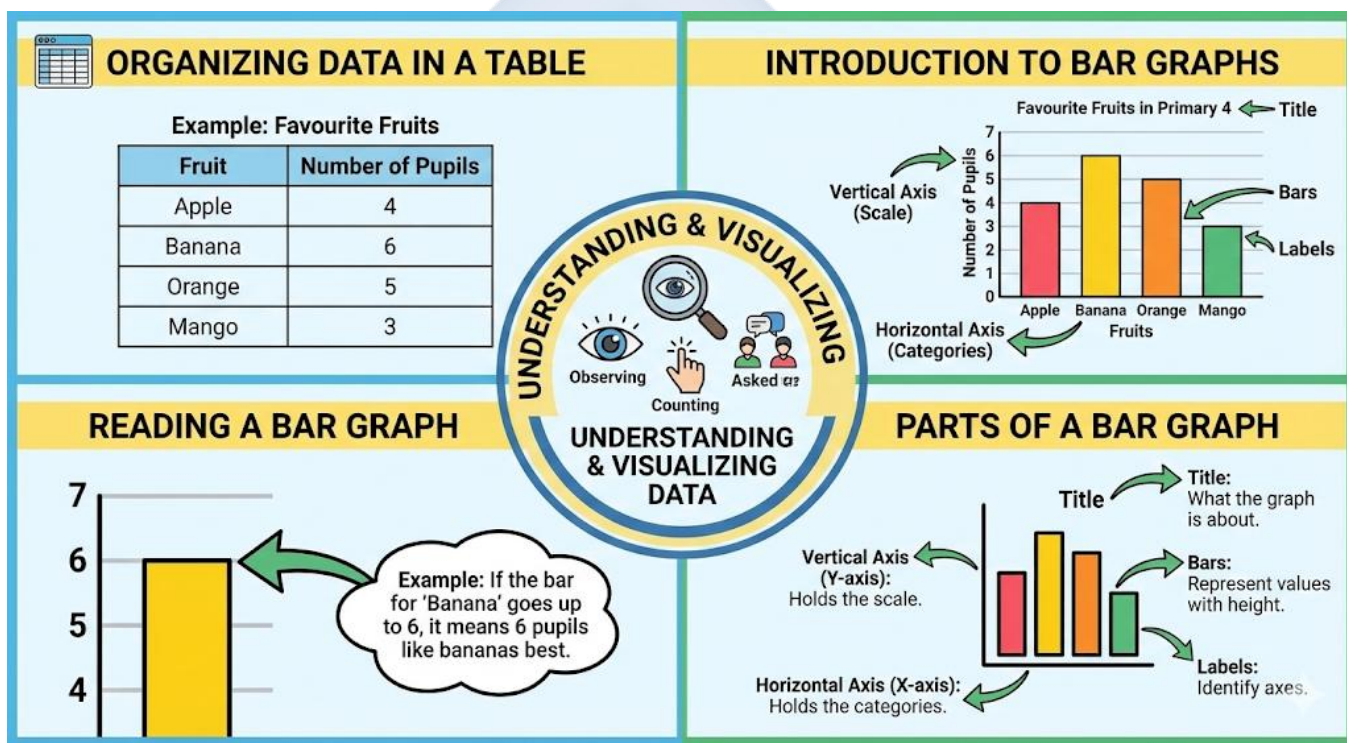
WEEK 9: BAR GRAPHS (PART 1)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Define data as information collected.
2. Organize data using a table.
3. Identify the parts of a bar graph (title, labels, bars, scale).
4. Interpret (read) information from a simple bar graph.

CONTENT



Understanding Data

Data is a collection of facts or information. We collect data by **observing, counting, or asking questions**. For example: "What is the favourite colour of pupils in our class?"

Organizing Data in a Table

A **table** helps us arrange data neatly so it's easy to read.

- *Example: Favourite Fruits*

Fruit	Number of Pupils
Apple	4
Banana	6
Orange	5
Mango	3

Introduction to Bar Graphs

A **bar graph** uses rectangular bars to represent numbers. The **height** or **length** of each bar shows the value for that item. Taller/longer bar = bigger number.

Parts of a Bar Graph:

1. **Title:** Tells us what the graph is about (e.g., "Favourite Fruits in Primary 4").
2. **Horizontal Axis (X-axis):** Usually holds the **categories** or names (e.g., names of fruits).
3. **Vertical Axis (Y-axis):** Usually holds the **scale** (numbers 0, 1, 2, 3...).
4. **Bars:** The rectangles whose height matches the number on the scale.
5. **Labels:** Words that identify what is on each axis (e.g., "Fruits" on X-axis, "Number of Pupils" on Y-axis).

Reading a Bar Graph To read a bar graph, find the bar for the item you are interested in. See how high it goes on the scale. That number is the value.

- *Example:* If the bar for "Banana" goes up to the number 6 on the scale, it means **6 pupils** like bananas best.

EVALUATION

1. What is data? Give an example.
2. Where is the title of a graph usually found?
3. If a bar in a graph is very short, does it represent a small number or a large number?
4. Look at a simple bar graph drawn by the teacher. Which fruit is the most popular? How can you tell?

5. What does the vertical axis of a bar graph usually show?

ASSIGNMENT

1. **Data Collection & Table:** Ask 5 friends in your neighbourhood for their favourite sport (Football, Running, Swimming, Other). Record your findings in a table.
2. **Drawing:** On a grid in your book, draw a single bar that represents the number 7. Label the scale from 0 to 10.
3. **Oral:** Explain to a family member why a bar graph can be more useful than just a list of numbers.



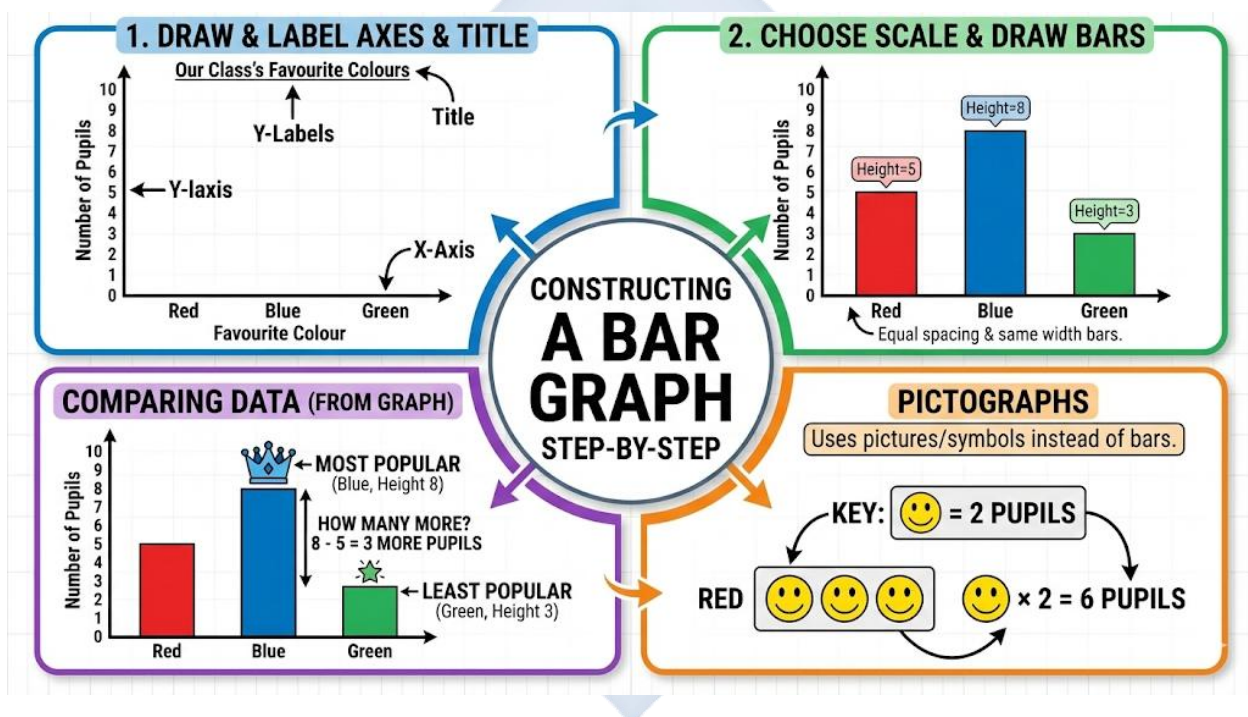
WEEK 10: BAR GRAPHS (PART 2)

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Construct a simple bar graph from given data.
2. Choose a simple scale (e.g., 1 unit = 1 item).
3. Compare data using the bars.
4. Interpret a pictograph.

CONTENT



Constructing a Bar Graph Step-by-Step

Let's make a bar graph for "Favourite Colours" using this data: Red: 5, Blue: 8, Green: 3.

1. **Draw the Axes:** Draw two perpendicular lines (an 'L' shape).
 - The vertical line is the **Y-axis** (for numbers).
 - The horizontal line is the **X-axis** (for colours).
2. **Add Labels and Title:**

- Label the X-axis "Favourite Colour".
 - Label the Y-axis "Number of Pupils".
 - Write the title at the top: "Our Class's Favourite Colours".
3. **Choose a Scale:** Decide what number each line on the Y-axis will represent. Since our biggest number is 8, we can mark lines from 0 to 10. Each line can represent **1 pupil**.
4. **Draw the Bars:**
- For "Red" (value 5), draw a bar above the "Red" label on the X-axis. Make the top of the bar reach the line for **5** on the Y-axis.
 - Do the same for "Blue" (bar height 8) and "Green" (bar height 3).
 - Leave equal spaces between bars. Make all bars the same width.

Comparing Data from a Bar Graph

A graph makes comparison easy!

- **Most Popular:** The **tallest bar** (Blue, height 8).
- **Least Popular:** The **shortest bar** (Green, height 3).
- **How many more?** "How many more pupils like Blue than Red?" Subtract: $8 - 5 = 3$ **more pupils**.

Pictographs

A **pictograph** uses **pictures or symbols** instead of bars to represent data. A **key** tells us what each picture stands for.

- *Example:* A smiley face ☺ = 2 pupils.
- If "Red" has 3 smiley faces, it means $3 \times 2 = 6$ **pupils** like red.

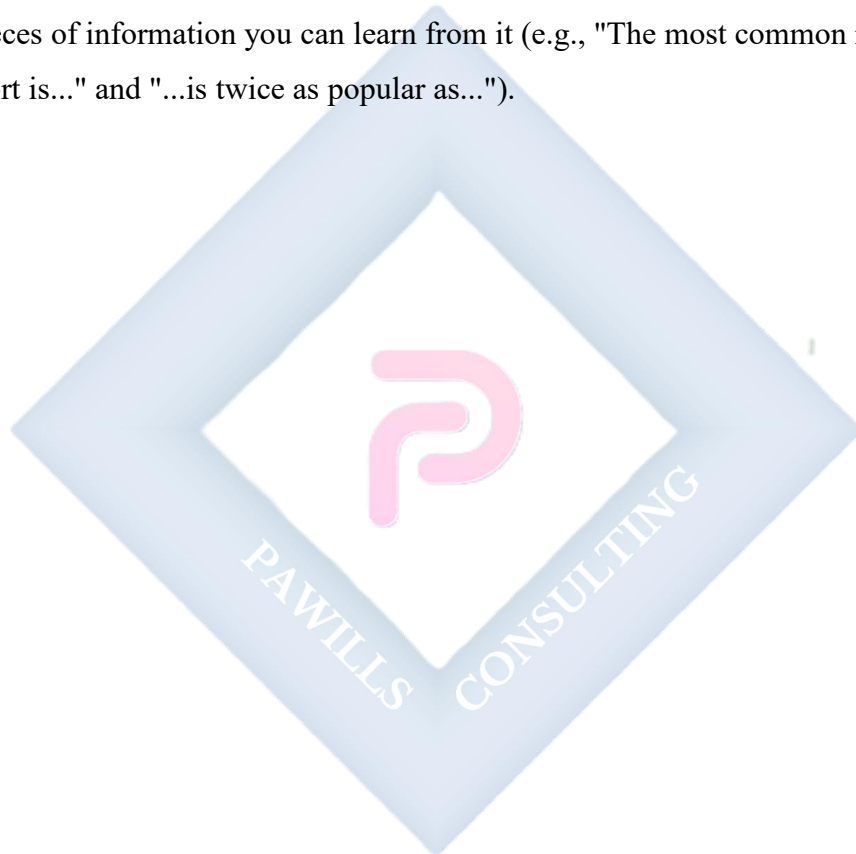
EVALUATION

1. Using the data (Cats: 4, Dogs: 6), draw a simple bar graph with labelled axes.
2. In the graph you drew, which pet is more popular?
3. What is a pictograph?
4. In a pictograph, if one star ★ represents 2 books, how many books do 4 stars represent?

5. Why is it often easier to compare numbers using a graph than a list?

ASSIGNMENT

1. **Graph Creation:** Use the data you collected in Week 9 (favourite sports from 5 friends). Create a bar graph to display it.
2. **Pictograph Drawing:** Draw a simple pictograph to show: 4 mangoes and 2 oranges. Let one circle symbol ○ represent 1 fruit. Include a key.
3. **Analysis:** Look at a bar graph in your textbook or prepared by the teacher. Write down two pieces of information you can learn from it (e.g., "The most common mode of transport is..." and "...is twice as popular as...").



WEEK 11: REVISION

OBJECTIVES

By the end of the lesson, pupils should be able to:

1. Recall key concepts from the entire term.
2. Solve mixed problems on Weight, Time, Area, Capacity, Shapes, and Graphs.

CONTENT

Comprehensive Review of Third Term

- **Weight & Capacity:** Understanding 'how heavy' vs. 'how much it holds'. Key conversions: **1000g = 1kg**, **1000ml = 1l**. Adding and subtracting these measures.
- **Time:** Reading clock faces (o'clock, half past, quarter to/past). a.m. vs. p.m. Calculating duration. Days, weeks, months.
- **Area:** Space inside a shape. Formulas: **Rectangle: $L \times W$** , **Square: $Side \times Side$** . Units: cm^2 , m^2 .
- **Shapes:**
 - **2D (Plane):** Flat, have sides & vertices (circle, triangle, square, rectangle, pentagon, hexagon). Lines of symmetry.
 - **3D (Solid):** Have faces, edges & vertices (cube, cuboid, sphere, cylinder, cone). You can hold them.
- **Graphs:** Data collection. Parts of a bar graph (title, axes, labels, bars, scale). Drawing and reading bar graphs and pictographs. Comparing data.

EVALUATION

- Oral quiz ("How many sides on a pentagon?") and written revision exercises covering the term's work (e.g., "A rectangle is 8cm by 5cm. Find its area.", "If a movie starts at 5:30 p.m. and ends at 7:00 p.m., how long is it?").

ASSIGNMENT

- Study all notes for the end-of-year examination.

- Create a one-page revision poster for one of the topics (e.g., "All About 3D Shapes" with drawings and properties, or "How to Read a Bar Graph" with an example).

